Tau Core: An Atemporal Morphological Readout Framework and a Conditional Minimum-Viable Physical Completion

Foundation Paper I

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Abstract

Tau Core asks whether observed four-dimensional physics can be represented as an observer-indexed, information-losing readout of a richer atemporal morphological object. We define the minimal architecture $s \mapsto \rho_\tau(s) \mapsto M_\tau \mapsto \mathrm{Desc}_O(M_\tau)$, enforce source-freeze, null-quotient and no-double-counting rules, and fix one minimum-viable completion. The common primitive-readout principle is

$$Y_O^{(a)} = Q_O^{(a)} \circ T^{(a)} \circ A_O[M_\tau],$$

where observer position, extent and relational access act before typed time, quantum, gravity, matter, radiation and gauge terminals, and stable sector-specific resolution determines operational classes. There a body-side carrier becomes an operational observer only together with a regular local rank-four descent and a nontrivial stable record/effect; the observer and its accessible four-dimensional world are co-readouts of that same body-conditioned solution. Thus neither a four-dimensional chart nor carrier support alone is observerhood. There a stabilized body is solved before terminal variation, while one once-counted functional supplies gravitational, gauge/current, quantum-effect and clock rows. Under explicit Lorentzian, regularity, state/effect and observer-carrier assumptions, this construction has compatible conditional limits: a semiclassical Einstein equation with renormalized quantum stress and the fixed-background Schrödinger–Born description.

The same completion yields the single-record resolution bound

$$\delta t_{\min,O} \geq \frac{2\hbar}{\Delta H} \arcsin \frac{\epsilon_O}{2} \geq \frac{\hbar \epsilon_O}{\Delta H},$$

where $x_A = \epsilon_O^2$ is an independently reconstructed dimensionless body-action separation. Uhlmann minimization in the declared common-Gram completion gives $x_A = x_Q = 2(1 - \sqrt{F})$ and $D_B = 2\arcsin(\epsilon_O/2)$. The proposed Tau-specific discriminator is this independently testable body–Uhlmann identity, not the standard quantum speed limit or its inverse-energy dependence.

The main source result is conditional but finite. Inside an enriched special dagger-Frobenius record completion, a complete source-owned record edge induces a faithful trace-preserving standard-form map; primitive-defect extensivity then forces the exact Trace–Gram identity on its endpoint pair and composable path closure. Endpoint-only descent is permitted only when relative holonomy is fixed in advance as Uhlmann-gauge or null under the complete endpoint descriptor. This is a sufficient internal realization theorem, not a derivation that complete edges exist physically. Record occupation, the finite identity and operational hold are independent required parts of a test. Unrestricted parent-law realization and physical occupation, laboratory occupation, arbitrary-state coverage, numerical constants, ultraviolet/Standard-Model completion and empirical confirmation remain open. The paper therefore presents an auditable conditional completion, not a replacement for

general relativity or quantum theory. This is the architectural first stage of a seven-stage foundation programme. Papers II and III analyze finite source sufficiency and record transport; Paper IV isolates physical parent-law realization; Paper V develops temporal descent; and Paper VI develops the conditional quantum terminal. Paper VII is issued as three focused parts: VII-A assembles gravity, matter and Abelian gauge terminals, VII-B owns the rank-seven P3 parent packet and its empirical frontier, and VII-C owns the scalar-15/type-II and functional-RG specialization. None serves as empirical evidence for the claims proved here.

Complete source-characteristic refinement. The phrase "the base adds redundancy" does not mean that the physical base carries no information. The complete source characteristic is

$$\mathfrak{c}_{B,s} = (q_B, \beta_B),$$

where q_B decodes the universal seed and β_B retains every independently physical base modulus. Only $\ker D\mathfrak{c}_{B,s}$ is true realization redundancy. Separately, an oriented multiplicity-free primitive seed grammar has the unique complete unital realization $\Lambda^\bullet V_s$, with the empty word as unit. These are conditional source-class theorems, not evidence that the unrestricted physical parent occupies the class or that β_B is already exhaustive.

Dependency and ownership box.
Inputs from earlier papers: none within the Tau Core paper sequence. The manuscript declares every imported mathematical and standard-physics ingredient at the point of use.
Results owned here: the atemporal seed–body–observer architecture; source freeze, null and body-first rules; the common primitive-readout law; the conditional minimum-viable completion; and the series-wide claim ledger. Technical source-signature, record-transport, parent-realization, temporal and quantum proofs are owned by the later papers. The Paper-VII triptych separates joint terminal assembly, the P3 parent packet and the scalar-15/type-II functional-RG specialization. Those downstream results are not additional results owned by Paper I.
Paper II further proves that the mixed relation ideal, separator multiplicity matching, relative gluing and faithful retained-block occupation are not independent switches once one coherent source representation/state datum $\mathfrak{J}_* = (\Pi_*, \varphi_*)$ is supplied. The enriched MVP adopts this joint datum only as a working completion. Its unrestricted parent-law realization and the completeness of the future terminal family remain open.

Observer co-descent convention. The observer index O first labels an occupied body-side carrier candidate, not a detached or already operational observer. Define

$$O_{OS}^{\mathrm{op}} = \big(B_O, \mathrm{Desc}_{OS}^{(4)}, Q_{OS}, E_O\big),$$

where $\mathrm{Desc}_{OS}^{(4)}$ is a regular local rank-four Lorentzian descent, Q_{OS} is a stable observer–source resolution map, and $E_O \neq 0$ is a stable record/effect. The carrier and its accessible four-dimensional world are co-readouts of one body-conditioned solution. Rank four without a record, or carrier support without rank four and a record, is insufficient. The descent is a quotient/restriction of the body-resident relation law, not a separate physical channel layer and not a quantum collapse of the parent.

Upstream regular-4D reconstruction boundary. Before any typed terminal is evaluated, the claim-safe preterminal chain is

$$(\mathcal{A}_{OS}^{\mathrm{acc}}[M_\tau], z_{OS}[J_{OS}^{\mathrm{acc}}]) \longrightarrow \Xi_{OS}^{\mathrm{cont}} \longrightarrow (\mathcal{A}_{OS}^{\mathrm{alg}}, D_{OS}, \leq_{OS}) \longrightarrow Z_{OS}^{(a)} \xrightarrow{Q_{OS}^{(a)}} Y_{OS}^{(a)}.$$

Thus Ξ_{OS}^{cont} is the shared smooth preterminal descriptor, whereas the finite operational record is $D_{OS}^{\mathrm{op}} = Q_{OS,\delta}(\Xi_{OS}^{\mathrm{cont}})$. Differential and Hessian statements below apply to the former; hard resolution cells belong to the latter. The shorthand $\mathfrak{C}_{OS} = (\mathcal{A}_{OS}^{\mathrm{alg}}, \mathfrak{d}_\tau, R_{OS}^{\mathfrak{d}}, \{\omega_{OS,x}\}, G_{\mathrm{occ}})$ is a mathematical representation of this post-body descent, not a new ontological layer. Here $\mathfrak{d}_\tau$ is a body-side derivation family defined before the observer restriction, and $\Phi_{OS,x} = \widetilde{\Phi}_{OS,x} \circ R_{OS}^{\mathfrak{d}}$ defines the observer-visible tangent candidate. Constant local rank four, integrability, Lorentz-conformal realizability, a compatible density, and agreement with the existing ROOT/M4TR four-dimensional carrier are separate open gates. Even if they hold, one obtains a local coframe atlas; a global tetrad and a fermionic spin lift require additional topology. Thus the operational packet above is a theorem target, not a derivation of spacetime.

Atemporal parent-realization terminology. Because the Tau parent is atemporal, phrases such as *Nature selects* do not denote an external agent or a later choice. Their precise meaning is the internal physical implication

$$(B_\tau^{\mathrm{phys}}, s_U, \mathcal{L}_{\mathrm{parent}}) \Longrightarrow \mathfrak{P},$$

together with occupation of the support of the resulting packet $\mathfrak{P}$. A theorem inside a declared completion proves only class-relative realization. It does not prove this unrestricted base–seed arrow, and agreement of finitely many terminal readouts cannot by itself establish ambient-source exhaustivity. Current-facing statements therefore use *parent-law realization* and *physical occupation*; any retained legacy selection language is shorthand for this same distinction.

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1 Purpose and claim boundary

This manuscript states the Tau Core foundation language and the strongest currently frozen physical completion in a form that can be criticized, reproduced, and rejected. It is self-contained: no result from another Tau Core manuscript, repository, or unpublished analysis is used as a premise or as evidence for any claim made here.

The claim boundary has two levels. At the foundational level, the paper defines an atemporal body/readout architecture and its audit discipline. At the completion level, it fixes one common functional and proves consequences conditional on an explicit assumption ledger. Neither level supplies empirical validation or a theorem that the physical base–seed source selects that completion. The main exclusions are summarized below and itemized again near the end of the manuscript.

> **Reviewer-facing claim box.** This paper claims (i) a formal body/readout architecture and failure discipline, and (ii) one *conditional* minimum-viable completion with compatible GR/optics and quantum/Born limits plus one frozen observer-resolution lower-bound discriminator. It does not claim empirical validation, uniqueness, unrestricted parent-law realization and physical occupation, a new force, dark-matter or dark-energy replacement, ultraviolet quantum gravity, or full Standard-Model recovery. Every conditional result below is void as a claim about the physical universe unless the common completion is independently realized and occupied or its predeclared discriminator survives testing.

Series position and notation

The three manuscripts separate architecture, source sufficiency and record transport so that a result at one level is not silently promoted at another. Figure 1 is navigational only; this manuscript remains self-contained in the sense stated above.

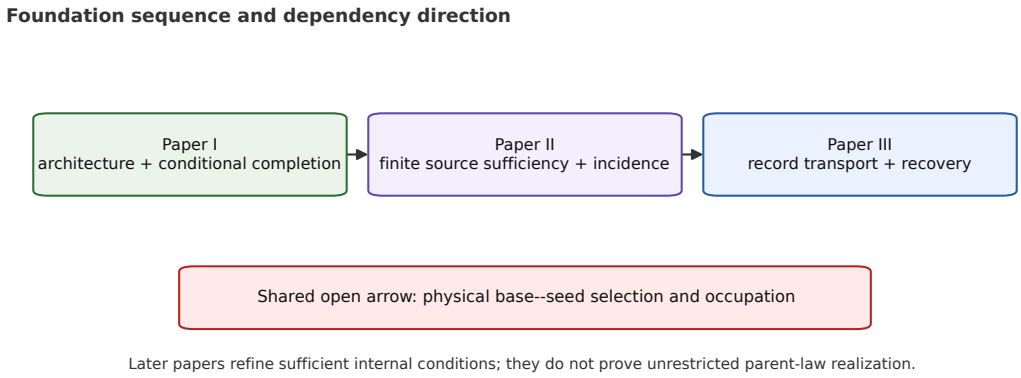


Figure 1: Dependency direction of the foundation sequence. The common red box is not closed by any of the three manuscripts.

Symbol	Series-wide meaning	Primary treatment
s	source/seed packet prior to observer descent	Paper I
M_τ	stabilized morphological response configuration (morphological body)	Paper I
Desc_O	observer-indexed physical descent/readout	Paper I
Π_{phys}	selected coherent source representation	Paper II
x_A	normalized finite source/body action separation	Papers I, III
x_Q	observer-carrier Uhlmann chord separation	Papers I, III
$\mathcal{C}, \mathcal{R}$	record map and recovery map	Paper III

The central idea can be written in one line:

$$s \longmapsto \rho_\tau(s) \longmapsto U_i^{M_\tau}(\rho_\tau(s)). \tag{1}$$

The first arrow is not time evolution. It is a parent-side response relation. The second arrow is a body-conditioned readout. The "readout" is not an extra physical substance or history store: in the completion developed below it is an observer-specific realization of ordering and compatibility already resident in the stabilized body. A physical branch appears only after the descent has supplied a codomain, a null policy, a closure test, and a validation boundary. When M_τ is fixed by the declared source class, the superscript is often suppressed and U_i is written for $U_i^{M_\tau}$. The notation O in such maps denotes the carrier leg before operational closure; it denotes an observer only after the rank-four, quantizer and record/effect clauses above are jointly occupied.

In a deterministic branch one may write

$$\rho_\tau : \mathcal{S} \to \mathfrak{R}_\tau.$$

In a relational branch the same symbol denotes either a relation

$$\rho_\tau \subseteq \mathcal{S} \times \mathfrak{R}_\tau$$

or, equivalently, a set-valued response rule

$$\rho_\tau : \mathcal{S} \to \mathcal{P}(\mathfrak{R}_\tau).$$

Such a branch must declare whether it uses the whole admissible response class or an endpoint-blind source-freeze rule selecting a representative.

Equivalently, the paper works with the following foundation package:

$$\mathfrak{T} = (B_\tau^{\mathrm{base}}, \mathcal{S}, \mathfrak{R}_\tau, \rho_\tau, M_\tau, \{U_i\}, \{N_i\}, \{J_i\}).$$

Here B_τ^{base} is the pre-readout carrier/support context, $\mathcal{S}$ is a seed/source domain, $\mathfrak{R}_\tau$ is a formal response domain, ρ_τ is a map, relation, or set-valued response rule, and M_τ is the stabilized morphological response configuration (short name: morphological body). The U_i are sector readouts, N_i are null or quotient policies, and J_i are optional relifts. This package is an organizing object for later branches (Fig. 2), not a proof that the unrestricted physical base–seed parent law realizes and occupies it.

In informal Tau Core discussions, the phrase *Tau Core base field* refers to the carrier/support role B_τ^{base}, not to M_τ. The seed is its base-readable relational/loading pattern, and their complete stable realized response is M_τ. The phrase "field" remains heuristic: this paper does not yet endow the base with ordinary spacetime dimension, time evolution, physical units, stress–energy, or empirical calibration.

MRC-DEF2 fixes the word "object" on this side of the architecture. A base or seed object is an object of a typed parent relational domain, not an ordinary material body already embedded

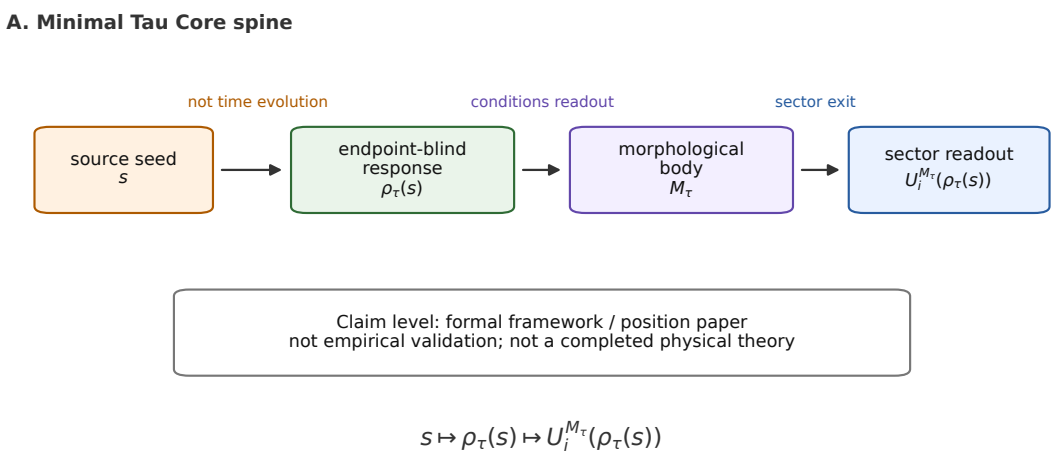


Figure 2: The minimal Tau Core spine used in Foundation Paper I. A parent-side seed s induces an endpoint-blind Tau response $\rho_\tau(s)$. A Tau-side morphological body M_τ conditions sector readouts U_i. The diagram is the weak foundation spine; the stronger conditional completion is shown in Fig. 5.

in the spacetime that the readouts are intended to recover. The base and seed are intended as pregeometric physical candidates, while their unrestricted realization and Nature occupation remain open. Ordinary particles, apparatuses and observers occur only downstream as persistent internal structures/readouts of M_τ; using one as the upstream exciter of the same body would be circular.

PGP-CON1: pregeometric-physics contract. The parent side is therefore treated as a separate typed research layer, not as a second substance or a completed physical theory. Its minimum candidate package is

$$\mathfrak{P}_\tau = (\mathcal{X}_B, \mathcal{X}_s, \mathcal{X}_M, \mathcal{C}_\tau, \mathcal{G}_\tau, \mathcal{L}_{\mathrm{pre}}, \mathcal{Q}_{\mathrm{pre}}),$$

where the entries type the base, seed, response space, simultaneous constraints, candidate symmetries, endpoint-blind parent-law class and quotient policy. The general law $\mathcal{L}_{\mathrm{pre}}$ need not be a scalar action. Formal nonemptiness of its response class, source-owned physical realization and nonzero Nature occupation are three different claims. Only a frozen stabilized body may enter the later observer–source descent; terminal recovery cannot be read backward to select this package. The canonical full contract and its adversarial controls are maintained in the Tau Core theory hub.

Method note. The manuscript was developed in an AI-assisted theoretical-research workflow. The AI workflow is not used as evidence for any physics claim; it is only part of the drafting and audit process.

This paper is therefore neither a phenomenology paper nor only a position paper. It is a foundation paper containing a conditional worked completion. It is useful only if that completion makes the program more falsifiable. A broad ontology that cannot be turned into frozen source rules, null tests, or endpoint-blind predictions is not useful as physics.

2 Foundation architecture and formal discipline

The paper has three jobs: define the body/readout architecture, exhibit one conditional physical completion, and state the exact physical-realization and empirical boundaries. The motivating

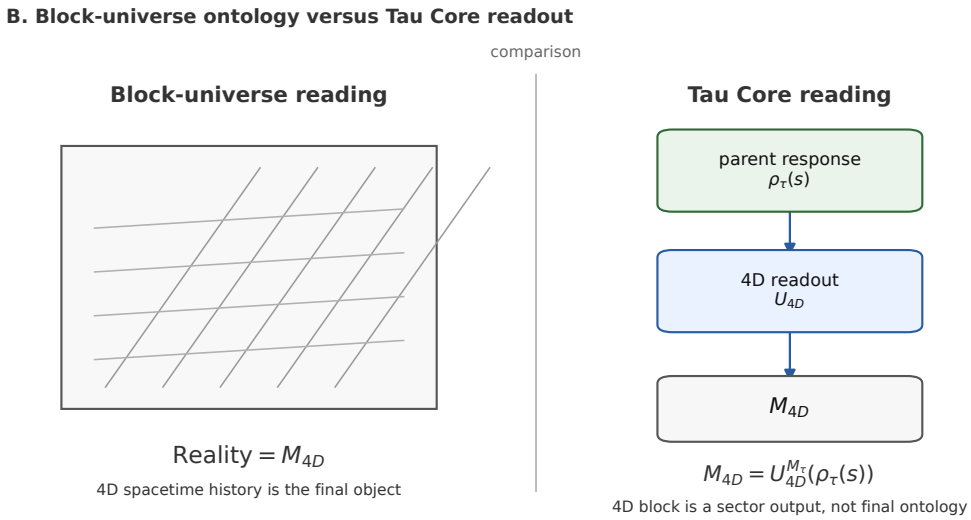


Figure 3: A four-dimensional block may be an effective final description for one readout while remaining a descended sector of the Tau parent. This is a positioning distinction, not a refutation of block-universe accounts.

question is whether time, spacetime, quantum state and other terminal descriptions can be observer-indexed readouts of a deeper response rather than mutually independent primitives. Related timeless, relational-time, reconstruction and constructor-theoretic programs show why this question is legitimate, but none is imported as evidence for Tau Core [9, 18, 7, 12, 6, 8].

The historical symbol τ names the parent side, not an additional parent clock. A stable four-dimensional block is therefore typed as

$$M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s)), \tag{2}$$

rather than assumed to be the final ontology. Ordinary spacetime field theory remains the effective theory on that descended leaf. This distinction has physical content only when another source-frozen readout imposes a constraint that is not already contained in the 4D endpoint.

The primitive vocabulary is finite. A seed $s \in \mathcal{S}$ is selected without endpoint leakage. The endpoint-blind response ρ_τ may be a map, relation or set-valued rule. Candidate realizations include a variational extremum, constraint solution, algebraic closure, probabilistic kernel, fixed point, or functorial construction; this paper does not identify those mathematically different mechanisms. The morphological body M_τ is the intrinsic structured object conditioning the readouts, not an observed 4D morphology label. A readout is a typed map

$$U_i : \mathcal{M}_\tau^{\mathrm{adm}} \times \mathfrak{R}_\tau \to \mathcal{O}_i$$

with a declared codomain, null policy and validation boundary. A relift $J_i : \mathcal{O}_i \dashrightarrow \mathfrak{R}_\tau$ is optional and need not be unique. If it is supplied, the coordinate-free obstruction is

$$\omega_i(r) = 0 \quad \Longleftrightarrow \quad J_i U_i(r)\, N_i\, r, \tag{3}$$

while $J_i U_i(r) - r$ is only a chart-level representative. These definitions do not themselves construct the physical body or explain a dark sector.

Every readout induces the equivalence relation

$$r \sim_i r' \quad \Longleftrightarrow \quad U_i(r) = U_i(r'). \tag{4}$$

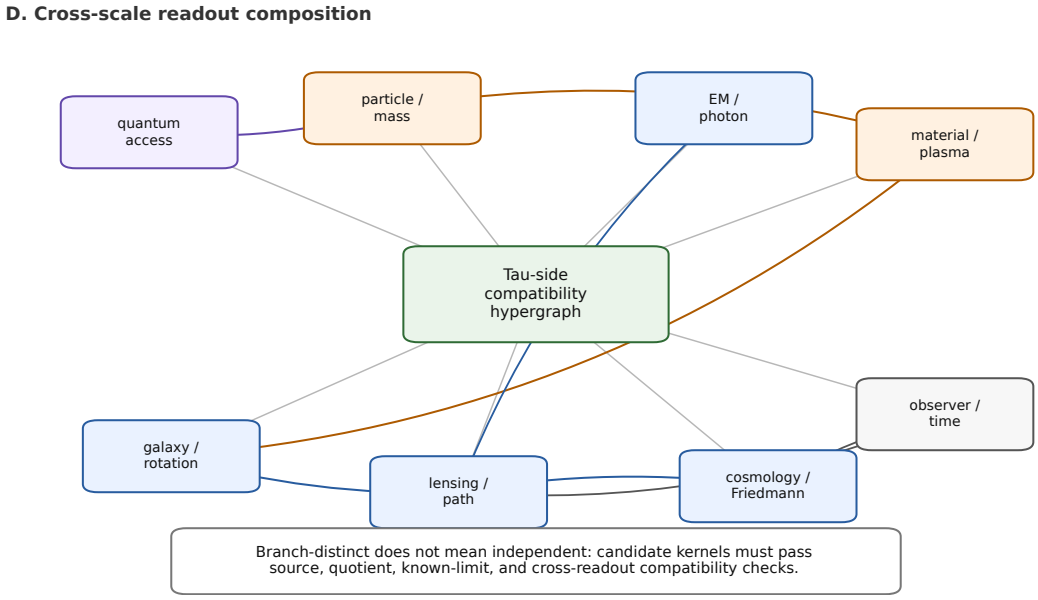


Figure 4: Readout families are branch-distinct but not presumed independent. Source, carrier, quotient and compatibility relations must be audited before separate terminal names are counted as separate information.

A joint readout has equivalence relation $\sim_i \cap \sim_j$ and therefore refines each component. Refinement is not statistical independence: several readouts may transmit the same parent degree of freedom. Tau Core accordingly requires a source-overlap ledger and forbids counting source-overlapping terminal terms as independent corrections unless their split or covariance is fixed upstream.

The weak-spine status is correspondingly strict: $\mathcal{S}$, $\mathfrak{R}_\tau$, ρ_τ, M_τ, U_i, N_i, J_i and ω_i are typed definitions or declared domains. The MVP below adds one stationary body, positive quotient Hessian and gravity/optics, quantum/Born and QOR leaves, but does not prove that the unrestricted physical parent law realizes or occupies them.

The terminal inventory is likewise explicit rather than unlimited. The spacetime/gravity family must recover metric, causal and lensing structure; observer-time must specify ordering, clock normalization and observer dependence; quantum access must recover state, effect, Born and no-signalling structure; matter/source must identify occupied modes and stress-energy; and the cosmological family must recover expansion, redshift and early-universe limits. Electromagnetic, gauge and composite astrophysical observables are later typed descendants of this common packet, not permission to introduce an independent correction for every anomaly. None of these names counts as a separate information source until overlap and factorization have been proved.

3 A conditional minimum-viable physical completion

The preceding architecture is deliberately weak: by itself it does not select a physical action, metric signature, quantum state, or observer. This section fixes one minimum-viable physical (MVP) object so that the paper contains an auditable theory chain rather than an indefinitely extensible list of possible readouts. The frozen chain includes a quantum observer-resolution (QOR) discriminator and is

$$\boxed{\begin{aligned}&\Gamma_{\mathrm{MVP}} \longrightarrow M_* \longrightarrow \mathrm{Desc}_O(M_*)\\ &\longrightarrow \{\mathrm{GR/optics}, \mathrm{quantum/Born}, \mathrm{QOR}\}.\end{aligned}} \tag{5}$$

Every arrow in Eq. (5) is conditional. The chain is internally composable and has a frozen discriminator, but unrestricted upstream parent-law realization and physical occupation remain open (Fig. 5).

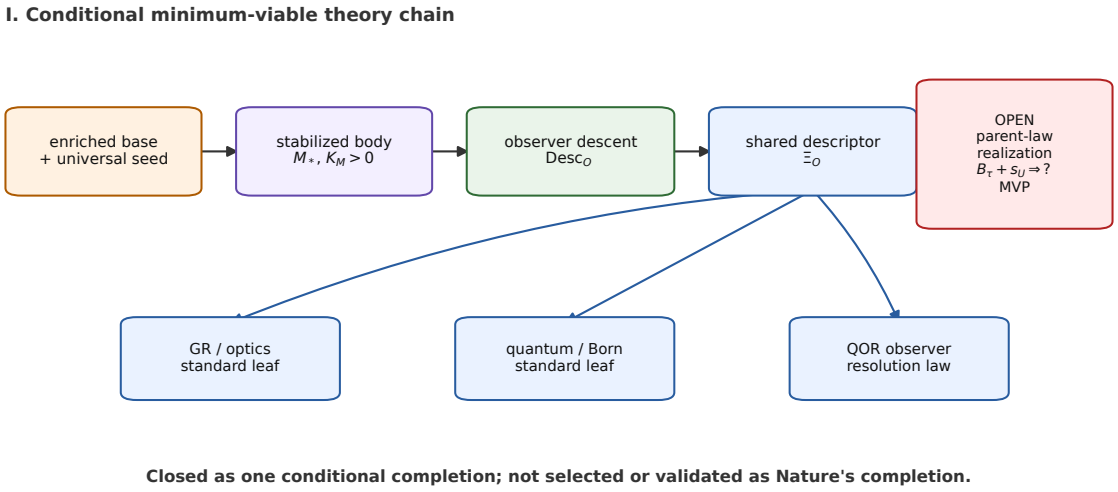


Figure 5: The conditional minimum-viable chain. A physically enriched base and universal seed are assumed to stabilize a body before observer descent. One shared descriptor then supplies the standard leaves and the QOR discriminator. The red box is the exact open step: internal realization and occupation of this completion by the unrestricted base–seed source law.

3.1 Frozen assumption ledger

The completion uses the following assumptions. They are deliberately separated from consequences.

Label	Assumption used in the MVP completion
A1	A physically enriched base and one universal seed admit a stabilized body M_* with a twice-differentiable body action.
A1-SHR	On the primary finite SHR branch, the seed owns a separating Alexandrov possibility family with at least one strict prerequisite cover; seed and body entries related by the realization null are two presentations of one pure deformation.
A1-USC	The universal seed owns every primitive ordering distinction of the occupied body response. A source-owned decoder $q\widehat{R} = I$ maps the complete stabilized body realization back to that seed order; a terminal-visible decoder-null ordering component would falsify this assumption.
A2	After quotienting only directions annihilated by every claimed terminal, the body Hessian K_M is positive on the retained tangent space.
A3	The body is solved first and frozen; terminal variations do not re-stabilize it or feed endpoint information into its definition.
A4	Coframe, connection/gauge, occupied matter, clock, and effect probes are once-counted terms or sources of one post-body generating functional.
A4-SRCMIN	The minimum-viable source ledger admits no independently occupied primitive carrier unless it has a separately typed, rank-novel recovery role. In particular, no independent base causal-work carrier is present in the MVP.
A5	The relevant descended leaf is orientable and Lorentzian; its gravitational term lies in the torsion-free Palatini/Einstein–Hilbert infrared class and is sufficiently regular for the variations used below.

Label	Assumption used in the MVP completion (continued)
A6	The quantum leaf admits positive normalized states and a complete effect family; the valuation on complete effects is additive and noncontextual.
A7	The field-theory leaf admits a locally covariant Hadamard prescription for the renormalized stress tensor and its local curvature ambiguities.
A8a-OCC	The physical completion occupies a nonzero source-owned recovery support, at least one record incidence, and a CP record instrument on that support. The finite pointer algebra and common Uhlmann amplitude bundle are then derived, not separately assumed.
A8b	The same post-body source that owns $x_O = \mathcal{A}_X^{PC}/\mathcal{A}_*$ selects a normalized amplitude connector whose horizontal Hilbert–Schmidt pullback equals the normalized body-action metric. On the explicit mixed-state completion it selects $V_{\mathrm{phys}} = V_{r(x_O)}$, with $r(x_O)^2 = x_O - x_O^2/4$ and $0 < x_O \leq 2$. In the enriched complete-record-edge completion, the connector and its finite identity are derived from the common trace/action character; physical occupation of that completion remains open.
A9	The quantum leaf uses the common action unit $\mathcal{A}_* = \hbar$, and self-adjoint transport obeys the Fubini–Study/Bures quantum-speed-limit bound.

A1–A4-SRCMIN, including A1-USC, define the working completion. A5–A9, including A8a-OCC, are standard-limit and prediction assumptions. None is claimed to follow from the old minimal Tau skeleton alone. In particular, positivity of a body Hessian does not select the Lorentzian leaf, the Born valuation, or a physical observer carrier. Section 3.10 proves a sufficient algebraic source theorem for A8b and then derives affiliation inside the declared primary SHR minimum-source completion. A8b is therefore a conditional theorem of that MVP packet, not a theorem of old-bare Tau or an unrestricted physical-realization result.

The physical motivation for A1-USC is source economy plus order consistency, not mathematical necessity. A universal seed is intended to be the compact owner of the primitive possibility and prerequisite distinctions that the base realizes redundantly as a body. If the realization produced an additional primitive order with no seed preimage, then two bodies agreeing on the universal seed could disagree on causal incidence; the seed would no longer be universal in the stated sense. A1-USC excludes that ambiguity and makes the body a realization of one source order. This motivation does not prove that the unrestricted physical base–seed law realizes and occupies such a seed packet: an order-generating base interaction is an admitted competing completion and is an explicit falsifier of A1-USC.

3.2 Body-first variational order

Let $\mathcal{A}_{\mathrm{body}}$ denote the atemporal body action. Stabilization means

$$D_M \mathcal{A}_{\mathrm{body}}(M_*) = 0, \qquad K_M = D_M^2 \mathcal{A}_{\mathrm{body}}\big|_{M_*} > 0 \quad \text{on } T_{M_*}\mathcal{M}/\mathcal{N}_{\mathrm{true}}. \tag{6}$$

Here $\mathcal{N}_{\mathrm{true}}$ contains only directions annihilated by every terminal retained in the MVP. A direction invisible to one terminal is not discarded if another terminal sees it.

The order in Eq. (6) matters. If z denotes all post-body terminal variables, linear response is solved in the ordered form

$$K_M\, \delta M = -j_M, \tag{7}$$

$$K_z\, \delta z = -j_z - C\, \delta M, \tag{8}$$

where $C = D_z D_M \Gamma|_{M_*}$ is an allowed body-to-terminal coupling. Equations (8) define a lower-triangular response problem. Placing M and z in an unconstrained simultaneous symmetric fit would reverse the generative direction and permit endpoint data to alter the body. The completion therefore forbids that operation unless a separately sourced post-body copy of the body variable is introduced and justified.

Proposition 1 (Complementary role reconstruction and local response)**.** *Suppose that the source family has a physically distinguished zero member, uses one source-independent stationarity, stability and selection law, and contains a stable sourced response* $[M_s] \neq [M_0]$*. Then the source contrast* $[\mathcal{I}_B(s;\cdot)]$ *is nonzero modulo body-independent constants. The frozen family therefore requires two noninterchangeable response roles: the zero-source law is the base role and the nonconstant contrast is the seed role. On a smooth regular fixed-base branch,*

$$K_B \dot{M}_0 = j_s, \qquad K_B = D_M^2 \mathcal{A}(0_B, M_0) \succ 0, \qquad j_s = -\, D_\lambda D_M \mathcal{A}(s(\lambda), M_0)|_{\lambda=0}\,. \tag{9}$$

If the base varies as well, the right-hand side is $j_s + j_B$*, so seed ownership cannot be inferred without an independent base freeze or reconstruction.*

This is a role theorem, not a two-substance theorem. The exact family

$$\mathcal{A}_0^{(\alpha)}(M) = \frac{1}{2}\langle M - \alpha d, K(M - \alpha d)\rangle, \qquad \mathcal{I}_1^{(\alpha)}(M) = (1-\alpha)\langle Kd, M\rangle, \tag{10}$$

with $K \succ 0$ and $d \neq 0$ has the same occupied minimizer $M = d$ and Hessian K for every α, but zero-source minimizer αd. Hence the occupied body and its Hessian do not identify the base/seed allocation. A physical zero-source comparison or equivalent intervention is indispensable; ontic separateness, universal-seed occupation and the homogeneity class of either role remain open.

3.3 Exact localization of seed-induced inhomogeneity

This subsection studies one declared branch; it does not settle the physical homogeneity class of either the base or the seed. The alternatives homogeneous-base/structured-seed, inhomogeneous base, mixed base–seed origin and spontaneous nonunique breaking are all admissible before the following premises are independently established.

The base–seed ownership split has one exact consequence in the quadratic unique-response branch. Let P_0 project onto the occurrence-homogeneous sector and set $Q = I - P_0$. Write the source-first body action as

$$\mathcal{A}(m; s) = \frac{1}{2}\langle m, K_B m\rangle - \langle b_0 + J_U(s), m\rangle, \tag{11}$$

where b_0 is the seed-independent base load. Assume

$$K_B > 0, \qquad [K_B, Q] = 0, \qquad Q b_0 = 0. \tag{12}$$

The last two conditions state operationally that the base is homogeneous and equivariant in this sector: it owns no transverse load and its response does not mix homogeneous and transverse representations.

The unique stationary solution and the base-only solution are

$$m_*(s) = K_B^{-1}\big(b_0 + J_U(s)\big), \qquad m_B = K_B^{-1} b_0. \tag{13}$$

Since K_B^{-1} commutes with Q,

$$\boxed{Q m_*(s) = Q\big(m_*(s) - m_B\big) = K_B^{-1} Q J_U(s).} \tag{14}$$

Invertibility on $\operatorname{ran} Q$ therefore gives

$$Q m_*(s) \neq 0 \quad \Longleftrightarrow \quad Q J_U(s) \neq 0. \tag{15}$$

Thus, under Eq. (12), the seed load owns the entire transverse mode while the base owns its susceptibility, smoothing and scale. The inhomogeneity originates in the seed load and is realized in the stabilized body; it is not already a four-dimensional density perturbation.

This attribution is conditional and its assumptions are necessary. Without $Qb_0 = 0$, the same nonconstant body can be generated by an inhomogeneous base load with $J_U = 0$. Without $[K_B, Q] = 0$, a positive stiffness can mix a homogeneous base load into a transverse response. Without uniqueness, a homogeneous invariant double-well action can select a nonconstant branch with zero seed load. Consequently a final body or terminal inhomogeneity alone does not prove the ownership premises or their physical realization by Nature.

Accordingly, Paper I does not claim that the universal seed owns all initial inhomogeneity or that the physical Tau base is homogeneous. Equation (14) is an exact conditional localization theorem, not a resolution of this origin fork.

3.3.1 Body-borne and readout-borne four-dimensional structure

Equation (14) localizes a transverse *body* mode; it does not by itself localize every contrast in an individual 4D readout. Let a_O denote any irreducible observer–source access datum retained after body stabilization and write the complete local 4D exit as

$$Y_{4,O} = \mathcal{R}_{4,O}(m_*, a_O). \tag{16}$$

For a readout-contrast projector $Q_{4,O}$, linearize about the base-only solution and set $\Delta m_* = K_B^{-1} Q J_U(s)$. The contrast separates as

$$\boxed{Q_{4,O}\Delta Y_{4,O} = Q_{4,O} D_m \mathcal{R}_{4,O}\, K_B^{-1} Q J_U(s) + Q_{4,O} D_a \mathcal{R}_{4,O}\, \Delta a_O.} \tag{17}$$

The first term is body-borne. It depends jointly on the seed forcing and the base susceptibility: removing the transverse seed load removes this term, whereas changing K_B changes its amplitude and profile. Thus the base is a constitutive or shaping cause even when it is not an independent symmetry-breaking source. The second term is readout-borne and must be derived from the source-frozen observer–source channel and stable resolution map; it is not another morphology seed.

A unique deterministic, observer-independent equivariant descent maps a homogeneous body to a homogeneous readout. An observer-relative correlated quantized descent can instead produce an inhomogeneous individual realization from a homogeneous body while preserving covariance of the observer family. That alternative is architecture-compatible, but its physical kernel, correlations, closure and cross-observer consistency are not derived here. Consequently the claim-safe causal ledger is: seed as transverse forcing, base as susceptibility and stabilization, and readout as downstream realization or additional observer-relative contrast. The observed 4D field alone does not identify their separate shares.

3.4 One common post-body functional

Three differently typed objects must be kept separate: the pre-body action used to solve M_*, the post-body microscopic functional before quantum or apparatus variables are integrated out, and the renormalized observer-indexed effective functional afterwards. Their lawful order is

$$\mathcal{A}_{\mathrm{pre}}[M; s, B] \xrightarrow{\mathrm{stat}_M} M_*, \qquad \Gamma_{\mathrm{micro,O}}[\Phi; M_*, e, A, \eta], \qquad \Gamma_{\mathrm{MVP}} \equiv \Gamma_{\mathrm{eff,O}}[M_*; e, A, \eta]. \tag{18}$$

The transferable body sector used in the first solve is not appended again downstream. An admissible once-counted microscopic ownership ledger is

$$\begin{aligned}\Gamma_{\mathrm{micro,O}} = {} & S^{\mathrm{bare}}_{\mathrm{grav}}[e] + S^{\mathrm{ret}}_{\mathrm{body}}[M_*; e, \Phi_{\mathrm{ret}}] \\ & + S^{\mathrm{bare}}_{q}[M_*; e, A, \Phi_q] + S_{\mathrm{src,O}}[M_*; e, A, \Phi_q, \Phi_{\mathrm{ret}}; \eta, \vartheta],\end{aligned} \tag{19}$$

where every primitive monomial or incidence has exactly one owner. Integrating the declared quantum and apparatus variables once defines

$$e^{iW_O^{\rm ren}[M_*;e,A,\Phi_{\rm ret},\eta,\vartheta]/\hbar} = \int_{\mathcal{C}_O} \mathcal{D}\Phi_q\, e^{i(S_q^{\rm bare}+S_{{\rm src},O})/\hbar}\Big|_{\rm ren} \tag{20}$$

and the effective packet

$$\boxed{\Gamma_{\rm MVP} = S_{\rm EH}[e] + S_{\rm loc}^{\rm ren}[e] + S_{\rm body}^{\rm ret}[M_*;e,\Phi_{\rm ret}] + W_O^{\rm ren}[M_*;e,A,\Phi_{\rm ret},\eta,\vartheta].} \tag{21}$$

Here $S_{\rm body}^{\rm ret}$ is the retained post-body representative; its role is to preserve the occupied body value and first variation without copying the already used pre-body envelope. If observer degrees of freedom and their source insertions are already included in $W_O^{\rm ren}$, there is no additional $\Gamma_{\rm obs}$ summand at the effective level. A separate observer term is admissible only when it owns a disjoint apparatus field domain.

This gives an exact de-duplication rule at quadratic order. For a Hessian block

$$K = \begin{pmatrix} K_x & B \\ B^\dagger & K_y \end{pmatrix}, \qquad K_{\rm eff} = K_x - BK_y^{-1}B^\dagger, \tag{22}$$

one may retain the full block or eliminate y and retain its Schur reduction, but not both. Propagators and Green response operators are inverses of second variations of Eq. (21); they are not additional action or stress summands. Likewise, for any allowed finite local covariant counterterm $C_{\rm loc}[e]$, the relocation

$$S_{\rm loc}^{\rm ren} \mapsto S_{\rm loc}^{\rm ren} + C_{\rm loc}, \qquad W_O^{\rm ren} \mapsto W_O^{\rm ren} - C_{\rm loc} \tag{23}$$

leaves $\Gamma_{\rm MVP}$ and its full variations unchanged. Adding the same counterterm to both terms would change the equation and is double counting. Equations (19)–(23) prove a once-counted effective and quadratic ledger once a microscopic ownership partition and renormalization prescription are supplied. They do not construct the full nonlinear microscopic Tau action or its ultraviolet completion.

For an occupied observer carrier O, define the typed shared descriptor

$$\Xi_O = {\rm Desc}_O\left(M_*, K_M, e, A, \psi, \vartheta, \{F_{O,a}\}\right). \tag{24}$$

More formally, on the adopted leaf

$${\rm Desc}_O : \mathcal{D}_{M_*}/\mathcal{N}_{\rm true} \longrightarrow \mathcal{X}_O,$$

where $\mathcal{D}_{M_*}$ is the typed post-body data bundle and $\mathcal{X}_O$ is the shared observer descriptor space. Quotient safety requires ${\rm Desc}_O(x) = {\rm Desc}_O(x')$ whenever $x \sim_{\rm true} x'$ under the retained common-null relation. The descent is the observer-specific realization of the ordering and compatibility law resident in the body. It is not a separate material layer, does not store a trajectory, and is not defined as a correction to a standard model. Standard physics and any future nonstandard component belong to the same full descent.

The terminal rows are generated by variations of the same packet:

$$\mathcal{E}_e = \frac{\delta\Gamma_{\rm MVP}}{\delta e}, \qquad \mathcal{E}_A = \frac{\delta\Gamma_{\rm MVP}}{\delta A}, \tag{25}$$

$$p(a|O,M_*) = \left.\frac{\delta W_O}{\delta\eta_a}\right|_{\eta=0}, \qquad \mathcal{E}_\vartheta = \frac{\delta\Gamma_{\rm MVP}}{\delta\vartheta}. \tag{26}$$

Common ownership fixes cross-sector covariance and prevents duplicate matter or Green-response terms (Fig. 6).

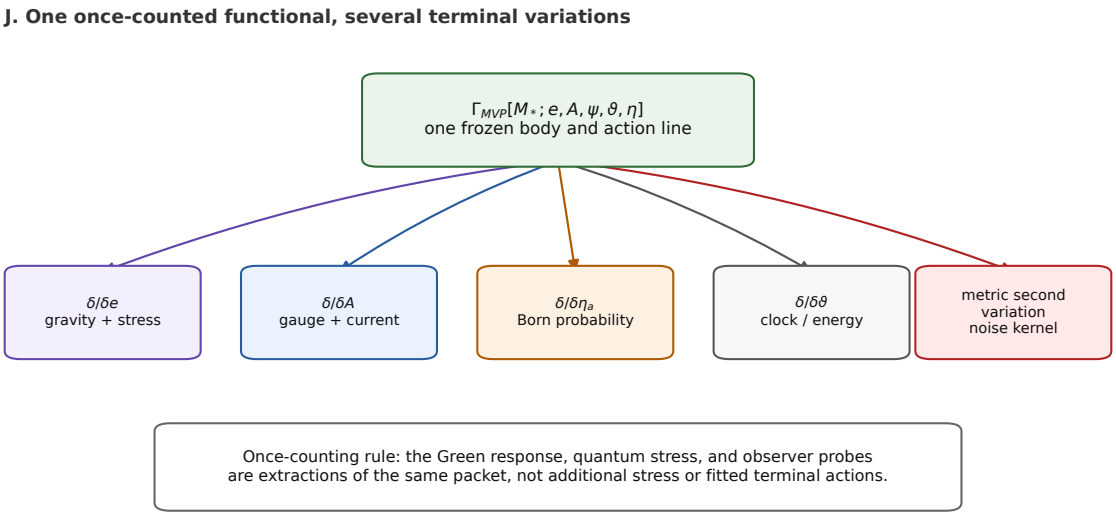


Figure 6: Terminal variations of one once-counted post-body functional. The diagram is a dependency map, not a claim that every displayed sector is already selected by the physical Tau parent.

Generalized internal observer–target response. Because the observer is internal to the occupied Parent/readout packet, a measurement may change the joint post-body state without reselecting the already stabilized body. Hold M_* fixed and let $\mathcal{A}^{M_*}_{\rm post}(x, o)$ be one regular common functional of a typed target coordinate x and observer/apparatus coordinate o. With

$$H_X = D_x^2 \mathcal{A}^{M_*}_{\rm post} > 0, \qquad C_{XO} = D_x D_o \mathcal{A}^{M_*}_{\rm post},$$

implicit stationarity and the terminal chain rule give

$$\boxed{D_o x_* = -H_X^{-1} C_{XO}}, \qquad D_o \Pi_a^{\rm occ} = D_o \Pi_a - D_x \Pi_a H_X^{-1} C_{XO}. \tag{27}$$

The two terminal terms are direct observer-context dependence and target-mediated backaction. A hard record requires a same-context clamped-target comparison rather than differentiating its quantizer; a null or same-cell terminal may hide a real target response, while a context-only terminal may change at $C_{XO} = 0$. Across several terminals, independent visibility is the rank of the stacked backaction rows, not the number of terminal names. This is a conditional common-functional theorem shared with ordinary interacting measurement models. Physical selection of C_{XO}, Nature occupation and a Tau-specific excess remain open.

3.5 Conditional gravity and optical leaf

On A1–A5 and A7, the low-curvature gravitational part of the completion has the effective form

$$\Gamma_g = \frac{1}{16\pi G_{\rm SHR}} \int d^4x \sqrt{-g}\,(R - 2\Lambda_{\rm ren}) + S^{\rm ren}_{\rm loc}[g] + W^{\rm ren}_O[g, A, \Phi_{\rm ret}, \eta, \vartheta; M_*], \tag{28}$$

where $S^{\rm ren}_{\rm loc}$ collects allowed local curvature counterterms and $W^{\rm ren}_O$ is the once-integrated quantum/observer functional. Coframe or metric stationarity gives

$$G_{\mu\nu} + \Lambda_{\rm ren} g_{\mu\nu} + H^{\rm loc,ren}_{\mu\nu} = 8\pi G_{\rm SHR} \left(T^{\rm body,ret}_{\mu\nu} + \langle T^{q+O}_{\mu\nu} \rangle_{\rm ren} \right). \tag{29}$$

Here $T_{\mu\nu}^{\mathrm{body,ret}}$ is the metric derivative of the retained body replacement term and excludes every degree of freedom integrated into W_O^{ren}. The second term is the metric derivative of that same effective functional and is not added as a separate microscopic action. Local covariance and diffeomorphism invariance give the corresponding Ward identity and conservation of the *complete* right-hand-side packet. The separate body and quantum pieces need not be conserved before their coupling and renormalization prescription are fixed. The structure of Eq. (29) is the standard semiclassical-gravity structure, including finite local curvature ambiguities [21, 13, 14, 15]; the Tau claim is its conditional common-source placement, not novelty of the equation.

If quantum stress, stress fluctuations, and local higher-curvature terms are negligible, Eq. (29) reduces to the Einstein equation. Its weak-field, slow-motion leaf gives the Poisson equation and Newtonian motion. For a source-free Abelian radiation field, the same Lorentzian leaf gives the eikonal conditions

$$k^\mu k_\mu = 0, \qquad k^\nu \nabla_\nu k^\mu = 0, \tag{30}$$

and the usual redshift and optical Jacobi equations. No independent lensing gain is introduced. These are conditional standard-limit recoveries; the completion does not predict numerical values of G_{SHR} or Λ_{ren}.

M4 metrological quotient. The occupied regular rank-four packet and the common P1 unit identities must be distinguished from a prediction of dimensionful decimal constants. With dimensionless local coframe coordinates, write

$$g_O = L_H^2 \widehat{g}_O, \qquad d\tau_O = T_H\, d\widehat{\tau}_O, \qquad c = \frac{L_H}{T_H}. \tag{31}$$

The common cone, gravity–action and action-period identities leave the single positive orbit

$$(L_H, T_H, \mathcal{A}_*) \longmapsto (\sigma L_H, \sigma T_H, \sigma^2 \mathcal{A}_*), \qquad \sigma > 0. \tag{32}$$

It changes the metric only conformally and preserves the cone, causal order, rapidities, proper-time ratios and all dimensionless clock–ruler terminals. One measured dimension-bearing anchor chooses an SI representative of this orbit without adding a terminal-specific gain. Conversely, the full σ-family is an identical-dimensionless-output counterfamily, so normalized morphology cannot predict the decimal SI value of that anchor. This conditional calibration closure does not fix dimensionless gauge or matter couplings and does not prove physical parent realization of the packet.

Source-arrow boundary for optical transport. A smooth optical family requires more than the pointwise metric and endpoint support. Two coefficient-free mechanisms are available once their source data are independently supplied. A body-owned connection ω_M gives holonomy

$$U_H = \mathcal{P} \exp\!\left(- \int_\gamma \rho_*(\omega_M) \right), \tag{33}$$

whereas a smooth constant-rank source-owned access projector gives canonical Kato transport

$$G_K = [\partial_x P_O, P_O], \qquad \partial_x U_K = G_K U_K. \tag{34}$$

Either mechanism yields the required optical naturality only when it represents the same occupied source arrows and transports both endpoint embeddings. A scalar $U(1)$ generator does not determine non-scalar projective mode transport, since

$$q_{\mathrm{pu}}(iaI_n) := iaI_n - \frac{\mathrm{tr}(iaI_n)}{n} I_n = 0. \tag{35}$$

Likewise, direct full-support access $P_O = I_n$ has $G_K = 0$. These are source-typing controls, not evidence for an additional optical effect. The current physical base–seed law has not been shown to realize either nontrivial source transport, and a terminal transfer path may not be inverted to define it.

3.6 Conditional quantum and Born leaf

At fixed descended metric, let ρ_O be a positive normalized state and $\{F_{O,a}\}$ a positive complete effect family,

$$\rho_O \geq 0, \quad \operatorname{Tr} \rho_O = 1, \qquad F_{O,a} \geq 0, \quad \sum_a F_{O,a} = I. \tag{36}$$

Under A6, complete additive noncontextual valuation yields the trace form

$$\boxed{p(a|O, M_*) = \operatorname{Tr}(\rho_O F_{O,a}).} \tag{37}$$

This is a Gleason-type representation result [10, 5]. Its precise domain assumptions matter: ordinary projective Gleason theory requires Hilbert dimension at least three, while generalized-effect versions use stronger measurement richness to cover the two-dimensional case. Tau Core does not infer Eq. (37) from information loss alone; positivity, completeness, and valuation structure are indispensable assumptions of the frozen completion.

At fixed background and in a regular positive-frequency sector, the same functional gives the Schrödinger description and locally covariant quantum field theory. A Hadamard prescription supplies $\langle T_{\mu\nu}^{q+O} \rangle_{\mathrm{ren}}$ in Eq. (29). Microcausality together with local completely positive trace-preserving operations gives operational no-signalling; entanglement changes joint statistics without creating a controllable superluminal signal. These are recovery conditions shared with standard algebraic QFT in curved spacetime [4, 13, 14], not novel predictions of the Tau architecture.

3.7 Compatibility of the standard limits

The gravity and quantum leaves are not two separately fitted models. They are limits of Eq. (21). Holding the descended metric fixed gives the quantum/Born leaf. Varying the coframe while retaining the renormalized quantum expectation gives Eq. (29). Suppressing quantum stress fluctuations and then taking the classical matter limit reaches the same classical geometry as taking the classical limit first and varying the corresponding classical matter action.

Proposition 2 (Conditional compatibility of the standard leaves)**.** *Under A1–A7, a common renormalization convention, and the stated regularity and limit–variation interchange assumptions, the fixed-background quantum limit and the small-fluctuation Einstein limit are compatible at the displayed semiclassical order. In the joint limit $\hbar \to 0$, suppressed stress noise, and negligible local higher-curvature terms, both routes reduce to classical matter on the same Einstein geometry.*

Conditional argument. Both routes are first variations of the same once-counted functional. The metric derivative of the renormalized matter term is the quantum stress in Eq. (29); replacing the state by its classical limit turns that derivative into classical matter stress. Because the body and renormalization convention are held fixed, no independent coefficient is introduced when the order of the two limits is exchanged. The stated regularity assumptions justify interchange of the limit and first variation. □

This is a completion-internal consistency result, not a proof of nonperturbative quantum gravity, metric quantization, or uniqueness (Fig. 7).

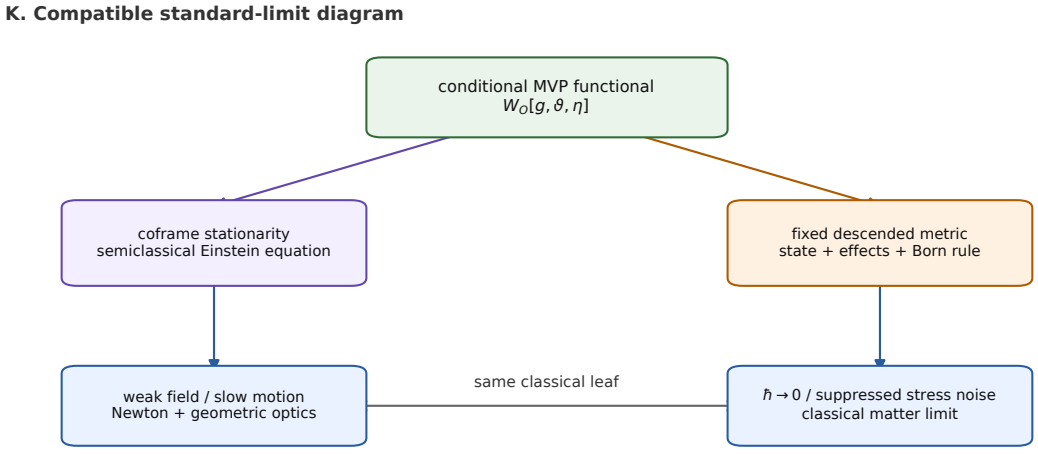


Figure 7: Compatible standard leaves of the frozen completion. Correct limiting behavior is necessary but not sufficient for physical realization or novelty.

3.8 QOR: a frozen observer-resolution discriminator

The MVP does not derive Lorentzian gravity or quantum state/effect structure from the weak Tau spine; it tests whether these adopted structures can be jointly organized as terminals of one frozen body-first completion. To do more than rewrite known equations, A8a-OCC together with A8b provides two source-frozen stable record prototypes $r_{O,0}$ and $r_{O,1}$ in the occupied observer carrier $\mathcal{C}_O$. Let $\iota_O : \mathcal{C}_O/N_O \to T_{M_*}\mathcal{M}/\mathcal{N}_{\mathrm{true}}$ be its source-declared linearized embedding into the retained body tangent space. The pullback of the positive body Hessian is

$$G_O = \iota_O^* K_M \iota_O, \qquad \mathcal{A}_X^{PC} = \langle \Delta r_O, G_O \Delta r_O \rangle, \qquad \Delta r_O = r_{O,1} - r_{O,0}. \tag{38}$$

The expression is coordinate independent and quotient-safe when ι_O descends to the displayed quotient. It has units of action because K_M is the body action Hessian. A8a-OCC and the support theorem below supply the common amplitude carrier, while A8b supplies the finite global completion of this local metric. Here "Gram amplitude" means an Uhlmann amplitude W satisfying $WW^\dagger = \rho$. Let normalized amplitudes $W_{O,i}$ satisfy $W_{O,i}W_{O,i}^\dagger = \rho_{O,i}$ and $\operatorname{Tr}\rho_{O,i} = 1$. The same inherited action metric is used on their right-unitary quotient, with no independent coefficient:

$$\frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} = \min_U \|W_{O,1} - W_{O,0}U\|_{\mathrm{HS}}^2, \qquad U^\dagger U = UU^\dagger = I. \tag{39}$$

The amplitudes are taken in a common sufficiently large purification space, so the finite-dimensional optimization can be written over right unitaries. If minimal purification spaces have unequal support, the equivalent partial-isometry formulation is understood; an infinite-dimensional extension would require separate domain and trace-class control.

Lemma 1 (Local metric of the common-Gram minimum lift)**.** *Assume constant rank in a neighborhood of q_0 and choose a smooth local horizontal gauge. Let $W(q)$ be a twice-differentiable normalized amplitude curve through $W(q_0)$, and let $\dot{W}_h$ be its horizontal tangent after minimizing the vertical right-unitary direction. Then*

$$\min_U \|W(q_0 + \delta q) - W(q_0)U\|_{\mathrm{HS}}^2 = \delta q^2 \|\dot{W}_h\|_{\mathrm{HS}}^2 + O(|\delta q|^3). \tag{40}$$

Under A8b, the pullback of this quotient metric is $G_O/\mathcal{A}_*$.

Proof. Write a nearby gauge representative as $W(q_0+\delta q)U(\delta q)^\dagger = W(q_0)+\delta q(\dot{W}-W(q_0)X)+O(\delta q^2)$, with $X^\dagger = -X$. Minimizing the quadratic coefficient over X removes the vertical tangent and leaves $\dot{W}_h$. Squaring the Hilbert–Schmidt norm gives Eq. (40). A8b identifies this horizontal quotient norm, with the common action unit restored, with the body-Hessian pullback in Eq. (38). The last identification is therefore explicit completion structure, not a consequence of Hessian positivity alone. □

A8b is the strongest physical identification in the QOR completion. The lemma proves the local metric of the Uhlmann quotient; it does not prove that the Tau body Hessian owns that metric. Body stationarity, Hessian positivity, endpoint blindness, and readout emergence do not by themselves imply A8b. Section 3.10 derives it only after the additional primary-SHR, pure-realization, recovery, and source-minimal premises have been declared.

Let $\mathcal{A}_* > 0$ be the common elementary action scale and define

$$\epsilon_O^2 = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} > 0. \tag{41}$$

Both numerator and denominator have the same units, so ϵ_O is dimensionless. The completion identifies the common action unit with $\hbar$ on the quantum leaf. It then declares a dimensionless quantum-state angle

$$D_Q(\rho_0,\rho_1) = \begin{cases} \arccos|\langle\psi_0|\psi_1\rangle|, & \rho_i = |\psi_i\rangle\langle\psi_i|, \\ \arccos\sqrt{F(\rho_0,\rho_1)}, & \text{mixed states}, \end{cases} \tag{42}$$

where

$$F(\rho,\sigma) = \left[\operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right]^2$$

is the Uhlmann fidelity. Uhlmann's minimum-lift identity [20] applied to Eq. (39) gives

$$\epsilon_O^2 = 2\left(1-\sqrt{F(\rho_{O,0},\rho_{O,1})}\right) = 4\sin^2\frac{D_Q(\rho_{O,0},\rho_{O,1})}{2}. \tag{43}$$

Consequently

$$D_Q(\rho_{O,0},\rho_{O,1}) = 2\arcsin\frac{\epsilon_O}{2} \geq \epsilon_O, \qquad 0<\epsilon_O\leq\sqrt{2}. \tag{44}$$

The final inequality uses $2\sin(x/2)\leq x$ on $0\leq x\leq\pi/2$. It is therefore a theorem of the common-Gram completion, not a separate distance assumption. Positivity of K_M alone would still be insufficient: the decisive input is the global quotient minimum-lift identity, which already includes all purification shortcuts.

For self-adjoint unitary transport, the Fubini–Study/Bures path obeys

$$D_Q(\rho(0),\rho(\delta t)) \leq \frac{1}{\hbar}\int_0^{\delta t}\Delta H(t)\,dt, \qquad \Delta H(t)^2 = \operatorname{Tr}(\rho H^2)-\operatorname{Tr}(\rho H)^2, \tag{45}$$

with mixed-state extensions expressed through the corresponding quantum Fisher information [16, 2, 19]. Equation (45) is restricted here to the declared self-adjoint unitary protocol. Open or noisy evolution requires its own quantum-Fisher-information path-speed bound and cannot inherit the same ΔH formula without an additional proof. For constant ΔH, combining Eqs. (44) and (45) gives the exact common-Gram bound

$$\boxed{\delta t_{\min,O} \geq \frac{2\hbar}{\Delta H}\arcsin\frac{\epsilon_O}{2} \geq \frac{\hbar\epsilon_O}{\Delta H}.} \tag{46}$$

The inverse-energy-spread form and the Uhlmann chord-to-angle identity are standard, not new Tau laws. The distinct Tau statement is narrower: the source-frozen morphological Gram

chord is physically identified with the observer carrier's Uhlmann-amplitude distance. At the same ΔH, two independently reconstructed observer carriers with different source-frozen ϵ_O have different forbidden single-record regions, provided the common-Gram carrier is physically occupied. The numerical prefactor is fixed by the angle convention in Eqs. (42)–(45). An ordering of *observed* onset thresholds follows only if the bound is saturated, or if independently controlled efficiencies establish the same slack above the bound. No detector gain may be fitted to enforce either conclusion.

For an empirical test, "independent reconstruction" therefore has a precise meaning: the carrier class, the two record prototypes, the embedding ι_O, and the pullback metric G_O must be fixed from source-side morphology and calibration data that exclude the timing endpoint. If these objects cannot be operationalized independently, Eq. (46) remains a conditional mathematical discriminator rather than a test-ready numerical prediction.

Here a *single stable record* means that one predeclared protocol run places a pointer variable in one of two disjoint decision regions with single-shot classification error at most η_{rec}, and that the assignment persists for a predeclared hold time τ_{hold}. No averaging across repeated preparations, tomographic reconstruction, or multi-shot parameter estimation is part of that record event. Those methods may calibrate the carrier or estimate sub-threshold changes, but they test a different operational question.

Finite carrier prototype and uncertainty ledger. A two-level carrier makes the full mathematical chain explicit without claiming physical occupation. The full qubit state carrier is the density- operator space $\mathcal{D}(\mathbb{C}^2)$; $\mathbb{CP}^1$ is only its pure- state projective submanifold. Restrict the prototype records to $\mathcal{C}_O^{\mathrm{pure}} = \mathbb{CP}^1$, choose $\rho_{O,0} = |0\rangle\langle 0|$, and set

$$|\psi_{O,1}\rangle = \cos\theta_O|0\rangle + \sin\theta_O|1\rangle, \qquad \rho_{O,1} = |\psi_{O,1}\rangle\langle\psi_{O,1}|.$$

Then $D_Q(\rho_{O,0}, \rho_{O,1}) = |\theta_O|$ for $0 \le |\theta_O| \le \pi/2$. The common-Gram identity gives the exact finite separation

$$\epsilon_O = 2\sin\frac{|\theta_O|}{2}, \qquad |\theta_O| = 2\arcsin\frac{\epsilon_O}{2} \ge \epsilon_O. \tag{47}$$

Thus the two-level example realizes the global theorem rather than assuming a second distance calibration. If a source-declared carrier coordinate q has tangent image $v_O = D\iota_O(\partial_q)$, set $g_O = \langle v_O, K_M v_O\rangle$. In a normal neighborhood,

$$\epsilon_O = |\Delta q|\sqrt{\frac{g_O}{\mathcal{A}_*}} + O(|\Delta q|^2). \tag{48}$$

The local Hessian estimate is not substituted for the exact finite minimum-lift distance outside that neighborhood.

To make "phase-calibrated quasistatic response" reproducible, choose before the timing experiment a closed calibration protocol $\mathcal{P}_{\mathrm{cal}}$ and define its unwrapped action phase by $\phi_{\mathrm{cal}}(q) = \mathcal{A}_{\mathrm{cal}}(q)/\hbar$. Near the source-declared reference q_0, the operational model is

$$\mathcal{A}_{\mathrm{cal}}(q) = \mathcal{A}_0 + f_0(q-q_0) + \frac{1}{2}g_O(q-q_0)^2 + O((q-q_0)^3). \tag{49}$$

For a symmetric predeclared step h_q, an endpoint-blind curvature estimator is

$$\frac{\widehat{g}_O}{\hbar} = \frac{\phi_{\mathrm{cal}}(q_0+h_q) - 2\phi_{\mathrm{cal}}(q_0) + \phi_{\mathrm{cal}}(q_0-h_q)}{h_q^2} + O(h_q^2). \tag{50}$$

Equivalently, if $f_q = \partial_q \mathcal{A}_{\mathrm{cal}}$ is measured, then $g_O = \partial_q f_q|_{q_0}$. This equality follows rather than being an independent calibration postulate when the protocol is the restriction of the same body action to a source-predeclared curve $M(q)$, with $M(q_0) = M_*$ and $v_O = M'(q_0)$:

$$\frac{d^2}{dq^2}\mathcal{A}_{\mathrm{body}}(M(q))\bigg|_{q_0} = K_M(v_O, v_O) + D_M\mathcal{A}_{\mathrm{body}}(M_*)[M''(q_0)] = g_O. \tag{51}$$

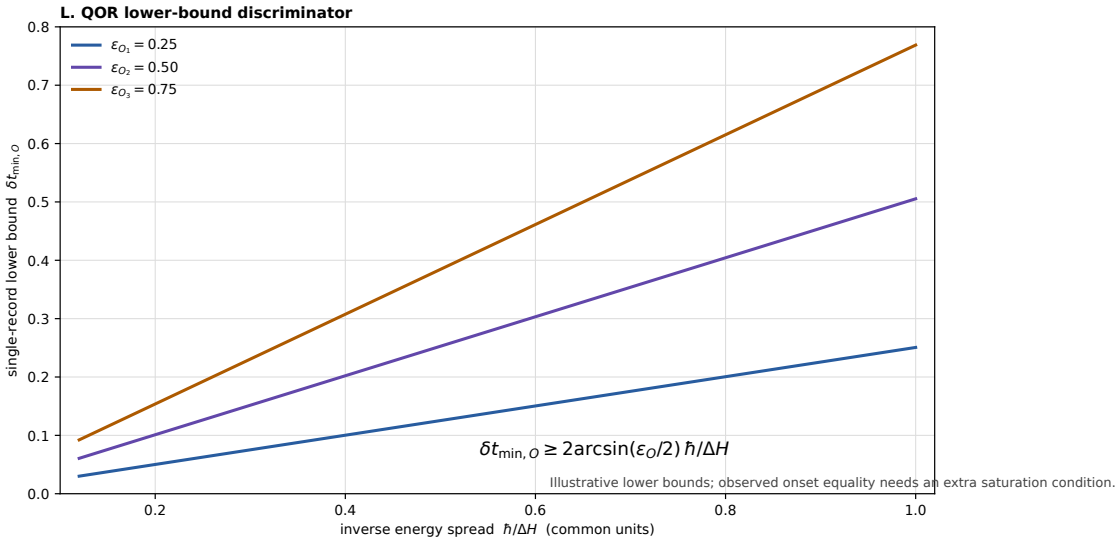


Figure 8: Illustration of Eq. (46). The slopes are not fitted or predicted numerical values. They show the ordering and ratio of *lower-bound* slopes after ϵ_O has been reconstructed independently. Equality of these lines with observed onset thresholds requires an additional saturation or controlled-efficiency condition.

The final equality uses body stationarity. The neighboring calibration configurations are source-prepared before the timing experiment; this is not terminal backreaction on the already frozen body. Failure of Eq. (51) rejects the proposed common-source carrier realization rather than authorizing a refit.

An endpoint-free calibration must therefore fix the carrier, q, q_0, h_q, phase-unwrapping rule, and $\mathcal{P}_{\mathrm{cal}}$ from non-timing device morphology; estimate $g_O/\mathcal{A}_*$ with Eq. (50); determine D_Q by state tomography; and determine ΔH by independent energy statistics. The common-Gram identity and Eq. (51) can then be tested before the timing endpoint is opened, avoiding conditional circularity. For a local reconstruction with independent Δq, g_O, and $\mathcal{A}_*$ estimates, first-order uncertainty propagation gives

$$\left(\frac{\sigma_\epsilon}{\epsilon_O}\right)^2 = \left(\frac{\sigma_{\Delta q}}{\Delta q}\right)^2 + \frac{1}{4}\left(\frac{\sigma_g}{g_O}\right)^2 + \frac{1}{4}\left(\frac{\sigma_{\mathcal{A}_*}}{\mathcal{A}_*}\right)^2, \tag{52}$$

or $\sigma_\epsilon^2 = J\Sigma J^T$ with the full calibration covariance. Metrologically $\hbar$ is exact in SI, while its identification with the parent action unit remains a physical assumption of the completion. Two carriers are comparable only under the same Hamiltonian generator and matched ΔH protocol; otherwise the predeclared comparison variable is $\Delta H\,\delta t/\hbar$, not raw time.

This prototype closes the type, normalization, distance bridge, and uncertainty bookkeeping. It also reduces pullback-stiffness reconstruction to the same-body action identity (51). It does not prove that the physical Tau source occupies this common-Gram carrier or supplies a laboratory realization of $\mathcal{C}_O$ and ι_O.

Equation (46) is not a universal time atom and permits continuous parameters. Aggregation may resolve sub-record changes. A test must therefore predeclare the carrier, record criterion, ΔH, aggregation rule, and estimator. Figure 8 shows bounds, not predicted onsets.

Minimal falsification protocol. The test is staged so that a failed carrier is rejected before timing data are used. On one predeclared carrier coordinate and record pair, define

$$\widehat{x}_A = \frac{\widehat{\mathcal{A}_X^{PC}}}{\mathcal{A}_*}, \qquad \widehat{x}_Q = 2\left(1 - \sqrt{\widehat{F}(\rho_{O,0}, \rho_{O,1})}\right). \tag{53}$$

Stage 1 reconstructs $\widehat{x}_A$ from the closed same-action curvature/work protocol and checks Eq. (51), without tomography or endpoint timing. Stage 2 reconstructs $\widehat{x}_Q$ by independent state tomography and tests the Tau-specific identity

$$\boxed{\widehat{x}_A \stackrel{?}{=} \widehat{x}_Q.} \tag{54}$$

For independent estimates, a minimal discrepancy statistic is

$$Z_A = \frac{\widehat{x}_Q - \widehat{x}_A}{\sqrt{\sigma_{x_Q}^2 + \sigma_{x_A}^2}}, \tag{55}$$

with the full declared covariance used when systematic errors are correlated. Defining either estimator from the other makes the test circular. Only a carrier passing both pre-timing tests proceeds to Stage 3, the QOR timing test. Thus failure of the common-Gram realization cannot be hidden by a timing fit.

Choose at least two observer carriers with independently reconstructed $\epsilon_{O_1} \neq \epsilon_{O_2}$, hold the Hamiltonian-spread and record protocol fixed, and estimate the single-record onset without using the timing endpoint in carrier selection. A reproducible stable record at $\delta t < 2\hbar \arcsin(\epsilon_O/2)/\Delta H$, after demonstrating that the one-record, common-Gram minimum-lift, source-freeze, and self-adjoint unitary-protocol conditions hold, rejects the completion. A cross-carrier onset ordering is an additional test only after saturation or equal-slack conditions have been independently justified. Failure of this completion does not by itself reject every possible Tau-style readout architecture.

3.9 Finite source criterion and carrier decision

The two parts of A8 can be reduced to finite source conditions rather than left as an undifferentiated bridge assumption. Let $K_* > 0$ be the body Hessian after the true-null quotient and let

$$R_{\mathrm{rec}}^{\mathrm{dedup}} : N_*^{\mathrm{phys}} \longrightarrow P_{\mathrm{rec}}\mathcal{H}_{\mathrm{rec}}$$

be the de-duplicated primitive recovery incidence. Define

$$G_*^{\mathrm{rec}} = R_{\mathrm{rec}}^{\mathrm{dedup}} K_*^{-1} (R_{\mathrm{rec}}^{\mathrm{dedup}})^\dagger, \qquad \rho_*^{\mathrm{rec}} = \frac{G_*^{\mathrm{rec}}}{\operatorname{Tr} G_*^{\mathrm{rec}}}. \tag{56}$$

Proposition 3 (Support-restricted common-amplitude theorem)**.** *Let $K_* > 0$ on a finite retained source tangent space and let $R_{\mathrm{rec}}^{\mathrm{dedup}} \neq 0$. Set*

$$P_* = \operatorname{supp} G_*^{\mathrm{rec}}. \tag{57}$$

Then ρ_*^{rec} is faithful on $P_*\mathcal{H}_{\mathrm{rec}}$, whether or not the recovery incidence is onto a larger declared ambient space. If one source-owned CP record instrument $\{\mathcal{I}_i\}$ acts on this occupied support and

$$\rho_i = \frac{\mathcal{I}_i(\rho_*^{\mathrm{rec}})}{\operatorname{Tr} \mathcal{I}_i(\rho_*^{\mathrm{rec}})} \neq 0, \tag{58}$$

then all ρ_i admit Uhlmann amplitudes in the same Hilbert–Schmidt carrier. A smooth local amplitude bundle exists on every constant-rank stratum.

Proof. For every y,

$$\langle y, G_*^{\mathrm{rec}} y\rangle = \left\| K_*^{-1/2} (R_{\mathrm{rec}}^{\mathrm{dedup}})^\dagger y \right\|^2,$$

so $G_*^{\mathrm{rec}} \succeq 0$ and $\ker G_*^{\mathrm{rec}} = \ker (R_{\mathrm{rec}}^{\mathrm{dedup}})^\dagger$. Restriction to $P_*\mathcal{H}_{\mathrm{rec}}$ is therefore strictly positive. Functional calculus gives the canonical amplitude $W_* = \sqrt{\rho_*^{\mathrm{rec}}}$, and polar decomposition gives

every minimal presentation as W_*U up to the usual partial-isometry extension. A CP map has positive output in the same support algebra, so every nonzero conditional state in Eq. (58) has a square-root amplitude in the same Hilbert–Schmidt carrier. Smooth functional calculus and a smooth horizontal gauge give the local bundle at constant rank. □

Thus common-amplitude existence is kinematic once the occupied support and record instrument are supplied. Ambient surjectivity and uniqueness among all stationary parent branches are not needed for this theorem; they would be stronger unrestricted-realization claims. What remains open in A8a-OCC is precisely whether the physical Tau parent occupies this support/instrument packet.

Record-domain and occurrence boundary. No additional apparatus ontology is needed to type one finite record once a positive effect is supplied. For any $0 \preceq E_O \preceq I$, introduce a two-state pointer initialized in $|0\rangle_P$ and set

$$\alpha_O = \arcsin\sqrt{E_O}, \qquad U_{E_O} = \exp(-i\alpha_O \otimes \sigma_y). \tag{59}$$

Projection onto the pointer basis gives

$$K_1 = {}_P\langle 1|U_{E_O}|0\rangle_P = \sqrt{E_O}, \qquad K_0 = {}_P\langle 0|U_{E_O}|0\rangle_P = \sqrt{I - E_O}, \tag{60}$$

up to an irrelevant branch phase. Thus one supplied occurrence produces the canonical square-root CP instrument without a fitted probability gain. Likewise, for any finite orthogonal pointer algebra and supplied record incidences T_j, the positive matching functional

$$\mathcal{A}_{\mathrm{rec}}(r_1, \dots, r_m \mid p) = \frac{1}{2}\sum_{j=1}^{m} \kappa_j \|r_j - T_j p\|^2, \qquad \kappa_j > 0, \tag{61}$$

has Hessian $\bigoplus_j \kappa_j I > 0$ and the unique structural minimum $r_j = T_j p$. The construction is natural in the finite pointer dimension; it adds neither a Born weight nor a new stiffness law.

Equations (59)– (61) prove availability and stability *conditional on an occurrence*. They do not select whether an observer–record edge is physically occupied. The same static carrier, effect, clock, and conserved current admit a zero-occurrence model and a one-occurrence model. Effect normalization fixes probability conditional on an interrogation, while carrier flux counts occupation rather than interrogations. A finite observer-time persistence interval is also stronger than positivity of the record Hessian: it requires a descended retention generator or direct operational verification. Hence A8a-OCC is reduced to a physical incidence statement, not a missing record-domain type.

A8b has two logically distinct defects. Let $\mathcal{C} : (N_*^{\mathrm{phys}}, K_*) \to (\mathcal{E}_*, h_*)$ be the proposed body-to-amplitude differential and let P_O be the h_*-orthogonal observer-access projector. Define

$$E_{\mathrm{src}} = \mathcal{C}^\dagger h_* \mathcal{C} - K_*, \qquad \Delta_O = \mathcal{C}^\dagger h_* (I - P_O)\mathcal{C} \succeq 0. \tag{62}$$

The actually descended metric is therefore

$$G_O = (P_O\mathcal{C})^\dagger h_* (P_O\mathcal{C}) = K_* + E_{\mathrm{src}} - \Delta_O. \tag{63}$$

The no-loss statement

$$\Delta_O = 0 \quad \Longleftrightarrow \quad \mathrm{im}\,\mathcal{C} \subseteq \mathrm{im}\,P_O \tag{64}$$

does not prove $E_{\mathrm{src}} = 0$. A8b follows without a fitted scale only if one and the same parent action supplies the normalized connector identities

$$\boxed{\mathcal{C}^\dagger h_* \mathcal{C} = K_*, \qquad P_O\mathcal{C} = \mathcal{C}.} \tag{65}$$

The first identity is source isometry; the second is complete access only on the selected record tangent, not on the whole parent body. Under both identities, quotienting the right-unitary vertical directions gives the Hilbert–Schmidt/Uhlmann metric with coefficient exactly $K_*/\mathcal{A}_*$. Multiplying $\mathcal{C}$ by a positive constant would change the first identity and is a different physical source incidence, not a coordinate convention.

Neither common support nor injective conformal descent selects this identity. Indeed, $\mathcal{C}_c = \sqrt{c}\,U$, with U an isometry and $c > 0$, has the same rank, complex structure, conformal cone, and normalized clock-line transport for every c, while its pullback metric changes by c. Consequently A8a-OCC and Hessian positivity do not imply A8b.

The consistency of A8b can nevertheless be exhibited without a fitted coefficient. Write

$$x_O = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*}. \tag{66}$$

For every $0 \le x_O \le 2$, define

$$r(x_O) = \sqrt{x_O - \frac{x_O^2}{4}}, \qquad \rho_\pm(x_O) = \frac{1}{2}\left(I \pm r(x_O)\sigma_z\right). \tag{67}$$

These are normalized positive qubit states and obey

$$2\left(1 - \sqrt{F(\rho_+, \rho_-)}\right) = x_O. \tag{68}$$

Thus A8b has a normalized mixed-state realization exactly on $0 \le x_O \le 2$; distinct records require $0 < x_O \le 2$. It is also a valid CP observer restriction: the depolarizing map $\mathcal{E}_r(\rho) = r\rho + (1-r)I/2$ sends an orthogonal qubit pair to Eq. (67) and admits a standard Stinespring isometry V_r.

This proves non-emptiness, not unrestricted parent-law realization and occupation. The current body/Gibbs and CP-descent laws admit every $r \in [0,1]$ and contain no term that chooses its value. The remaining physical statement is therefore the finite source identity

$$\boxed{V_{\text{phys}} = V_{r(x_O)}, \qquad r(x_O)^2 = x_O - \frac{x_O^2}{4}, \qquad 0 < x_O \le 2.} \tag{69}$$

It must be a second restriction of the same post-body functional that owns x_O. Choosing r from the timing endpoint or measured record overlap would be terminal fitting. Hence A8b is a mathematically closed conditional completion, while its physical source selection remains open.

Proposition 4 (Record occurrence and finite A8b are independent)**.** *Relative to assumptions A1–A9 without an additional law coupling occurrence degree to the amplitude connector, physical record occurrence and the finite A8b identity are logically independent.*

Proof. Fix $x_O \in (0,2]$ and define two propositions

$$\mathsf{O}: \; n_{\text{rec,O}} \ge 1, \qquad \mathsf{B}: \; 2\big(1 - \sqrt{F(\rho_+, \rho_-)}\big) = x_O, \tag{70}$$

where $n_{\text{rec,O}}$ counts source-owned record incidences in the declared observer support. The depolarizing Stinespring family used above is valid for every $r \in [0,1]$, and its symmetric output pair has squared Uhlmann chord

$$y(r) = 2\left(1 - \sqrt{1 - r^2}\right). \tag{71}$$

The value $r_*(x_O)$ in Eq. (67) satisfies $y(r_*) = x_O$. Choose any $r_0 \in (0,1]$ with $r_0 \ne r_*$ and $y(r_0) \ne x_O$. The models $(n_{\text{rec,O}}, r) = (1, r_*)$ and $(1, r_0)$ have the same body ratio, finite pointer

type, occurrence degree, and positive record-matching Hessian, but only the first satisfies B. Therefore $\mathsf{O} \not\Rightarrow \mathsf{B}$.

Conversely, hold the carrier geometry fixed at $r = r_*$ and compare $(n_{\mathrm{rec,O}}, r) = (1, r_*)$ with $(0, r_*)$. The finite metric identity holds in both admissible carrier completions, while only the first contains a physical record event. Thus $\mathsf{B} \not\Rightarrow \mathsf{O}$. Repeating the zero/one occurrence choice at r_0 realizes all four truth assignments of (O, B), proving independence. The zero-occurrence A8b case is a structural law on an available carrier, not a measured record. □

The proposition is relative to the current source axioms, not a prohibition on deeper unification. A richer post-body source law may select occurrence and the connector jointly. In the present completion, however, a realized QOR trial requires the conjunction

$$\boxed{n_{\mathrm{rec,O}} \geq 1 \quad \wedge \quad x_A = x_Q \quad \wedge \quad \mathsf{H}_{\mathrm{hold}}.} \tag{72}$$

Here $\mathsf{H}_{\mathrm{hold}}$ denotes the independently predeclared single-record retention condition. Occupation supplies an event, A8b supplies its finite morphology/Bures calibration, and the hold condition makes the event operationally readable; none substitutes for another.

3.10 Finite source closure: foundation-level summary

This subsection records only the source-level consequences needed by the foundation argument. The detailed complete-edge, holonomy and irreversible CP analysis and the detailed finite-signature, marginal and incidence proofs are technically separable. Their common message is that internal representation closure can be proved under explicit finite source data, while physical realization and occupation of those data remain open.

Blockwise spectral-to-amplitude transfer. On an occupied recovery decomposition, one source-owned connector with a single inherited action unit can transport orthogonal spectral blocks to orthogonal Uhlmann-amplitude blocks. Local metric agreement is insufficient for finite equality; the finite response must additionally be source-owned and extensive on its admitted direct-sum domain.

Sufficient internal realization of finite A8b. Inside the declared enriched special dagger-Frobenius packet, a complete source-owned record edge is faithfully trace preserving. Its standard-form lift is isometric, and primitive-defect extensivity forces

$$\frac{\mathcal{X}_s(e)}{\mathcal{A}_*} = d^2_{U,\mathrm{chord}}(\rho_u, \rho_v). \tag{73}$$

This is a sufficient internal realization theorem, not a proof that physical record edges are reversible or occupied.

Pathwise A8b and the holonomy boundary. Composable complete edges preserve Eq. (73) pathwise. Equal endpoints may be collapsed only when relative holonomy is Uhlmann-gauge or null under a source-frozen claim-complete descriptor. Descriptor-visible holonomy remains physical path information. A generic irreversible CPTP relation instead requires a certified recovery on its frozen occupied domain; commuting records alone cannot identify reversibility.

Proposition 5 (Primary-SHR recovery differential generates affiliation)**.** *Within the declared source-minimal primary-SHR realization, a source-owned nonzero nilpotent recovery differential and its Hodge–Dirac operator are affiliated with the finite recovery algebra. With the common-action amplitude connector this is sufficient for finite A8b.*

Proof sketch. The reconstructed seed boundary descends to the physical quotient, its differential squares to zero, and the differential and adjoint belong to the finite recovery algebra. Their self-adjoint sum is therefore affiliated. The statement remains conditional on the declared source-minimal realization and connector. □

Proposition 6 (Universal-seed rich-body recovery)**.** *If every occupied base–seed response stabilized before descent is included in one complete body, the body action is basic on the true-null quotient, and the universal seed owns the complete ordering presentation, then the recovery differential and affiliated Hodge–Dirac construction extend to the full body without requiring minimum body dimension.*

Proof sketch. The full realization defect vanishes under action basicness. Complete seed-owned order confines any chain mismatch to decoder-vertical redundancy; true-null descent removes only terminal-invisible directions. A visible decoder-null ordering component would falsify the premise and require a richer seed or an order-generating base law. □

3.10.1 Exact open step: physical parent-law realization

The internal results do not close the physical arrow:

$$B_\tau^{\mathrm{phys}} + s_U \quad \not\Longrightarrow_{\text{current derivation}} \quad \Gamma_{\mathrm{MVP}} + \text{state/effect completion} + \text{QOR carrier}. \tag{74}$$

The open packet includes the universal seed, complete body/order premises, occupied record support, same-source identity $x_A = x_Q$, coherent joint source map and observer hold. No downstream consistency theorem proves that the unrestricted physical parent law occupies these inputs.

3.10.2 Finite source-signature result

The finite source layer has one compact conclusion. For a finite source algebra $\mathcal{A}_s$, faithful state ω_s, and complete typed role packet, let Σ_s denote the multiplication, involution, state and distinguished-incidence data. Then:

Proposition 7 (Finite source-signature representation uniqueness and joint-incidence boundary)**.** *Two minimal cyclic finite realizations of the same complete faithful typed signature are unitarily equivalent by the unique typed unitary mapping one cyclic vector to the other. This proves representation uniqueness after complete source ownership; it does not supply that ownership. Separate sector marginals do not determine the joint source, and even all proper marginals in the four-qubit Pauli control leave the 81-dimensional weight-four subspace unresolved. Conversely, one source-owned onto joint incidence with positive stiffness produces a faithful joint Gram state.*

Proposition 8 (Sector marginals do not select the joint source)**.** *Faithful marginal states do not generally determine their joint source state.*

Proposition 9 (All proper marginals remain incomplete)**.** *In the four-qubit Pauli control, all proper marginals span 174 of 255 traceless directions and leave 81 genuine four-body directions unresolved.*

Proposition 10 (One onto joint incidence is sufficient)**.** *If $K_s \succ 0$ and the source-owned incidence $R_J : \mathcal{E}_J \to \mathcal{J}_s$ is onto, then the normalized Gram state $\rho_J \propto R_J K_s^{-1} R_J^\dagger$ is faithful and fixes every finite mixed moment.*

The proofs are finite: GNS gives typed cyclic uniqueness, the family $(I_{16} + \epsilon\sigma_z^{\otimes 4})/16$ proves the proper-marginal no-go, and surjectivity makes $R_J^\dagger$ injective. The once-owned primitive roles therefore compress to one coherent physical target

$$\Pi_{\text{phys}} : \mathcal{A}_{\text{prim}} \longrightarrow \mathcal{B}(\mathcal{H}_{\text{post}}), \tag{75}$$

whose kernel selects the relation ideal and whose quotient image selects the representation. Its unrestricted physical occupation remains open. The full proofs, scope conditions and representation-specific dimension audit are technically separable from this foundation statement.

3.10.3 Source-generated records and atlas summary

Proposition 11 (Dagger unfolding supplies the record-copy map)**.** *Inside the enriched special dagger-Frobenius completion, the adjoint unfolding map $\Delta = \mu^\dagger$ with $\mu\Delta = I$ supplies an isometric record-copy mechanism without a free gain.*

Proof sketch. Specialness gives $\Delta^\dagger\Delta = I$. The resulting copy map is therefore normalized in the inherited source metric. The unchanged narrow parent does not contain or select this two-leg carrier. □

At fixed occupied roots and resolution, closure under the frozen incidences gives an inclusion-minimal observer atlas. GNS, edge and atlas constructions organize source-owned roles; they create none. Missing occurrence, hold, finite action or microscopic ownership remains a physical blocker.

3.11 Dimensional and claim-status audit

The principal quantities used by the completion have the following roles.

Quantity	Dimensions	Status
M_*, K_M	branch-dependent body coordinate and action Hessian	assumed stabilized; positive after true-null quotient
G_{SHR}	$L^3M^{-1}T^{-2}$ in SI	not numerically predicted
Λ_{ren}	L^{-2}	renormalized infrared parameter; not predicted
$\mathcal{A}_X^{PC}, \mathcal{A}_*$	action	common units required
$x_O = \mathcal{A}_X^{PC}/\mathcal{A}_*$	dimensionless	body-action record ratio; A8b requires $0 < x_O \leq 2$ for distinct records
D_Q	dimensionless angle	adopted Fubini–Study/Bures distinguishability metric
ϵ_O	dimensionless	source-frozen Gram chord; $D_Q = 2\arcsin(\epsilon_O/2)$
ΔH	energy	protocol-frozen energy spread
$\delta t_{\min,O}$	time	conditional single-record lower bound
$\eta_{\text{rec}}, \tau_{\text{hold}}$	dimensionless probability, time	predeclared single-shot error ceiling and record-persistence interval

The resulting claim ledger is finite:

Statement	Status in this paper
Support-restricted common Uhlmann-amplitude bundle	derived once the occupied support and CP record instrument are supplied; physical occupation open

Statement	Status in this paper (continued)
A8b mixed-state realization and $0 < x_O \leq 2$ range	derived consistency result; finite A8b follows from affiliation plus the explicit source-owned block connector, while unrestricted physical occupation remains open
Complete-record-edge finite A8b	derived inside the enriched special dagger-Frobenius completion: the edge map is faithfully trace preserving and primitive extensivity fixes the exact Trace–Gram law; arbitrary-state, physical-parent and laboratory occupation remain open
Pathwise finite A8b	derived on the occupied connected record groupoid; endpoint-only descent holds only for descriptor-null or Uhlmann-gauge holonomy, while visible holonomy remains typed path information
Complete-edge existence	derived conditionally as a nonempty occupied ROOT edge orbit inside the enriched source-irredundant packet; unrestricted physical-parent occupation, laboratory occupation and arbitrary-state coverage remain open
Reversible versus irreversible occupation	provably indistinguishable on a commuting record algebra; a coherent off-diagonal witness or independently verified recovery map is necessary
Frozen holonomy null policy	defined by the complete source-frozen endpoint descriptor; central or path automorphisms are not null by default
Exhaustive rich-body extension	complete stabilized base response has zero complete-body realization defect; universal-seed chain descent preserves the affiliated recovery construction without minimum body dimension
Record occurrence versus finite A8b	logically independent relative to the current source axioms; a physical QOR trial requires occurrence, A8b, and operational hold jointly
Finite source signature and joint incidence	typed cyclic representation is unique after the packet is supplied; marginals remain incomplete, while one onto incidence is sufficient for a faithful joint Gram state
Dagger-Frobenius record unfolding	conditional theorem with no free record gain; the traced two-leg source and source-irredundant branch are constructed inside the enriched packet, while unrestricted branch realization and occupation remain open
Primitive occupied observer atlas	unique at fixed record roots and resolution; physical occupation and observer count open
Once-counted effective and quadratic ledger	Hessian, Schur elimination and duplicate-source controls structurally derived; physical filling by one full microscopic Tau action remains open
Semiclassical Einstein/optics leaf	conditional standard-limit recovery
Trace-form Born leaf on fixed or classical-dynamic geometry	conditional on the state/effect valuation and covariant-transport assumptions
Born structure on a quantized metric	compatible only on a supplied constrained joint geometry–matter algebra; current construction open
Compatibility of classical limits	conditional consistency result
Pre-timing identity $x_A = x_Q$	Tau-specific conditional discriminator; physical source value and carrier occupation open
QOR inequality (46)	standard Bures-angle quantum speed limit after A8b; downstream consistency/falsification stage, not an extra Tau delay
Unique selection from the physical base and seed	exact finite criterion derived, but unique selection from the current narrow source reduct is disproved by an inequivalent energy-sharing same-reduct completion
Numerical constants, UV completion, full Standard Model	open

Statement	Status in this paper (continued)
Empirical support for the Tau-specific common-source identification	open

4 Formal safeguards, postulates, and finite controls

The parent/readout notation is scientifically empty if it can be reduced to $\mathfrak{R}_\tau = \mathcal{O}$, $\rho_\tau(s) = O$, and $U = \mathrm{id}$. A promoted branch must therefore impose, before endpoint inspection, at least one nontrivial source relation, quotient-safe null, shared-source constraint, known-limit recovery, wrong-family exclusion, or other consequence that could fail. Known-limit recovery alone is not enough because an existing endpoint theory can always be harmlessly relabelled.

For readout U_i, let N_i be its declared null relation. Quotient safety means

$$rN_ir' \Longrightarrow U_i(r) = U_i(r'), \qquad U_i = \bar{U}_i \circ q_i, \quad q_i : \mathfrak{R}_\tau \to \mathfrak{R}_\tau/N_i. \tag{76}$$

The converse need not hold: additional identifications constitute readout loss. A stable observed object class may be represented by a basin $\mathcal{B}_O = \{r : R_{4D}(r) \simeq O\}$; this is a persistence scaffold, not a particle-stability or decay theorem. Protected source identity must not be collapsed unless that loss was declared as null structure.

Endpoint blindness operationalizes the same rule. Before scoring a target, one must freeze the source or proxy, readout family, null/quotient and overlap policies, forbidden endpoint fields, controls, failure conditions, and a dated freeze artifact. A residual used to choose the source class that later "explains" it is endpoint leakage, not a Tau prediction.

The seven unproved foundation postulates are: (P1) an atemporal Tau substrate; (P2) endpoint-blind seed/base response stabilizing a body; (P3) endpoint-blind source and kernel selection; (P4) complete physical descriptions as typed readouts, not residual-only corrections; (P5) effective variation only after a stable readout ordering or clock is available; (P6) a hidden defect only when relift fails modulo the declared null; and (P7) observational tensions as possible readout/closure defects only after controls and held-out tests. These postulates define the candidate framework; the MVP realizes one conditional branch but does not derive the postulates from the physical base–seed law.

A finite control makes the quotient discipline explicit. Let $r = (a, b, c) \in \mathbb{R}^3$,

$$U_A(r) = (a, b), \quad U_B(r) = (a, c), \qquad J_A(x, y) = (x, y, 0), \quad J_B(x, z) = (x, 0, z).$$

Then $h_A = (0, 0, -c)$ and $h_B = (0, -b, 0)$: hiddenness is readout-relative. Replacing U_A after inspection by $U_A^{\mathrm{bad}}(a, b, c) = (a, b + \alpha c)$ while retaining the old null is inconsistent. Likewise, a source rule $c = f_{\mathrm{src}}(a, b; \theta)$ is admissible only if f_{src} and θ were frozen upstream.

5 Atemporal body and effective readout-time

Let $r = \rho_\tau(s)$ be atemporal and suppose $U_i(r) = o_i(t_i)$. The derivative do_i/dt_i is meaningful inside that readout without being a parent-time derivative. Another readout of the same response may return a static constraint. There is no contradiction: time is the stable ordering used by one terminal description, not an external clock that evolves the parent.

For a finite witness, take $r = (a_0, \ldots, a_n, c)$, $U_{\mathrm{seq}}(r) = (a_0, \ldots, a_n)$, and $U_{\mathrm{sum}}(r) = \sum_k a_k$. The first supports the ordered difference $\Delta a_k = a_{k+1} - a_k$; the second loses that order. The parent record has not evolved. The readout has exposed an ordering.

The morphological body M_τ is the structured source object conditioning these readouts, not a visible galaxy shape or catalogue label:

$$O_i = U_i^{M_\tau}(\rho_\tau(s)). \tag{77}$$

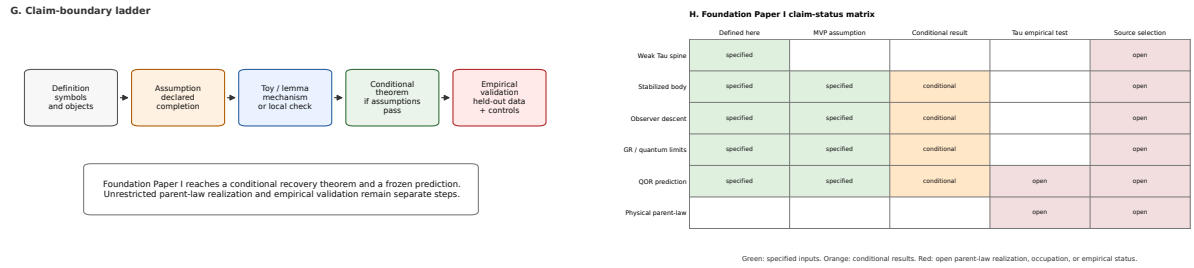


Figure 9: The claim ladder and status matrix. Definitions, postulates, conditional theorems, toys, numerical evidence and validation are distinct; the present paper reaches conditional standard leaves and a discriminator, not unrestricted parent-law realization, physical occupation or empirical validation.

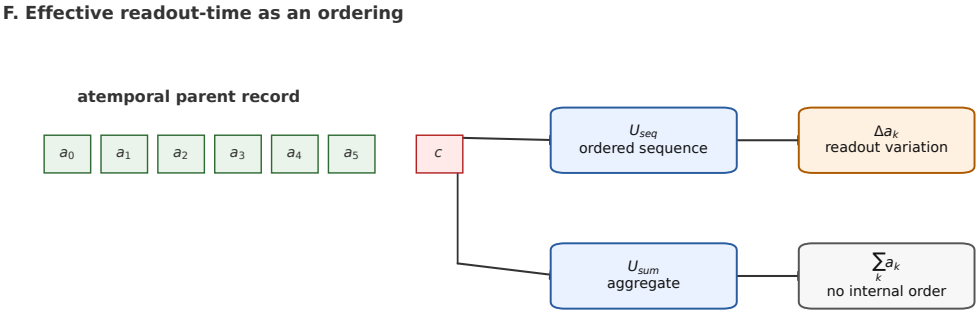


Figure 10: Effective readout-time in a finite toy. A sequence readout exposes ordering and finite differences; an aggregate readout of the same parent record does not.

Images, rotation curves, lensing maps, cosmic-web descriptors and measurement contexts are projected handles only. They can underdetermine global incidence, order, path geometry, closure and null structure. Body selection, kernel family and source overlaps must therefore be frozen before endpoint scoring, and two terminals using the same source coordinate may not be counted independently without an upstream split.

The resulting morphology-conditioned primitive-readout principle is

$$Y_O^{(a)} = Q_O^{(a)}\left(T^{(a)} A_O[M_\tau]\right), \qquad a \in \mathfrak{P}. \tag{78}$$

Here A_O contains occupied observer position, extent, relational or light-cone access, protected nulls and accessible modes; $T^{(a)}$ is the continuous typed precursor; and $Q_O^{(a)}$ is the stable source-derived resolution map. Equality under $Q_O^{(a)}$, rather than a generally nontransitive distance threshold, defines operational readout classes.

The "unfolding" available to an observer is structural rather than temporal. It consists of accessible rank, inherited response spectrum, closure depth, compatible redundancy, protected nulls and retained phase/holonomy. It is not evolution in a hidden parent clock. Different observer supports can expose different primitive-readout profiles while remaining restrictions of one body. Their overlap maps must commute with descent; arbitrary observer-specific correction functions are excluded.

Before the finite quantizers act, let

$$J_O^{(a)} = D(T^{(a)} \circ A_O)_{M_\tau}.$$

The stacked accessible rank obeys

$$\operatorname{rank} \begin{bmatrix} J_O^{(1)} \\ \vdots \\ J_O^{(n)} \end{bmatrix} \leq \operatorname{rank} A_O. \tag{79}$$

Thus different terminal names do not establish independent parent information. This is a conditional composition and rank theorem after the body/access packet is supplied; the paper does not derive physical base–seed realization and occupation of the complete packet or prove that the working primitive list is exhaustive.

Within the frozen complete eight-block source signature, the downstream packet is not further independent selection data. Body stationarity supplies M_τ; occupied common representation and incidence supply A_O and the joint typed precursor whose projections are the $T^{(a)}$; stable record partitions supply the $Q_O^{(a)}$; and the finite-metric and action-unit blocks remove terminal-specific rescaling. Hence the many apparent downstream selection arrows compress, conditionally, to the single upstream question of whether the physical base–seed parent selects and occupies that complete signature and observer support. This is an internal source-assembly result, not a proof of physical occupation.

There is, however, an exact sufficient occupation criterion. If the realized universal-seed response is cyclic for a faithful complete typed source representation and the base map annihilates no generated direction, its operation-generated orbit reaches every source block; declared operation closure then occupies the full packet. The seed itself need not contain one independent component per block. In a positive quadratic source chart this becomes $p_i K_{\mathrm{src}}^{-1} j_s \neq 0$ for every block. The present general base–seed axioms do not force this cyclic faithful exhaustivity: an all-block source and a one-block-omitted source can share the same narrow body reduct. Finite terminal agreement likewise cannot exclude a terminally invisible positive direct-sum extension.

Within the declared source-irredundant MVP carrier class, present redundant stable records supply a stronger branch-selection fact. The protected PUSC source is cyclic and faithful on its working parent, and the unique Frobenius base unfolding maps an occupied primitive alternative to the nonzero pair $e_* \otimes e_*$. The complete dependency ledger then reaches all eight source blocks. Thus actual record existence forces complete packet occupation inside the declared MVP, while leaving open whether the ambient physical parent contains a rank-novel independent carrier outside that class.

Observer time is correspondingly sector-typed:

$$t_{\mathrm{obs}}^S = U_{\mathrm{time}}^S(\rho_\tau(s), M_\tau, \mathcal{O}_S, \gamma_S). \tag{80}$$

The MVP adds an oriented clock descent and the conditional QOR law, but does not prove occupation of that clock carrier. Proper-time recovery and any nonrelativistic observer correction require a noncircular branch test. Future-labelled components of the parent record do not imply backward causation; the temporal order belongs to the descended readout.

The downstream temporal paper now closes the regular local interface needed by this architecture. On the adopted common P3 handoff, the linearized time terminal factors uniquely through the shared grade-one descriptor and preserves its rank and kernel. A protected temporal covector and positive contact response still select the clock; the P3 map alone does not. On the neutral restriction, matched morphology metric, observer access, temporal quantizer and calibration reproduce the standard GR proper-time line element exactly. Observer-relative rate, conditional morphological clock curvature and an observed Tau-over-GR anomaly remain

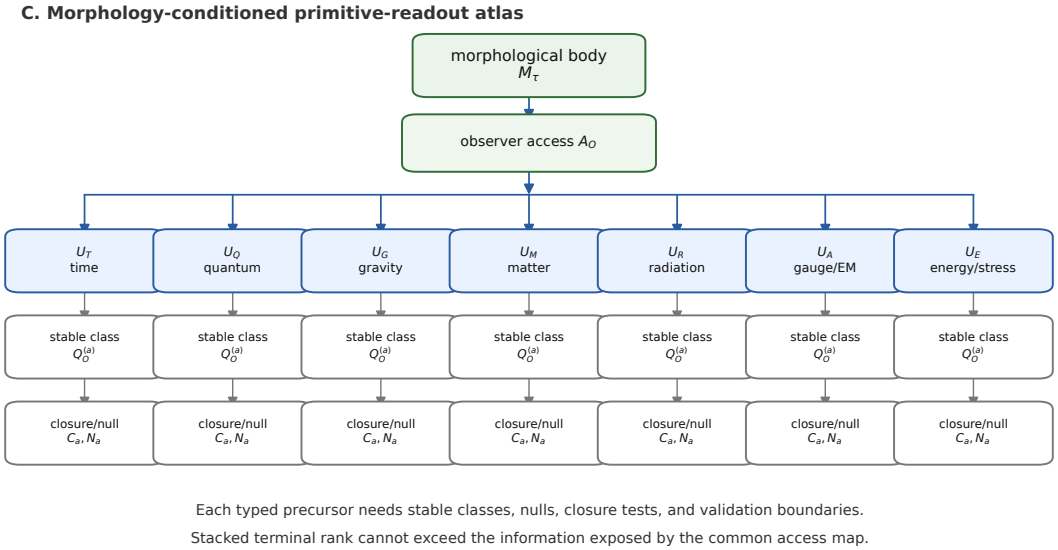


Figure 11: One morphological body and observer-access stage can support several typed primitive readout families. Their stable classes, nulls, closure tests and ownership must be audited before they are counted as independent information.

three distinct claims. The last is tested, not assumed, by a source-frozen path-pair ratio against a complete GR and propagation control. It also identifies the temporal branch as a fibre of integrable order enhancements over temporalizable atemporal carriers. The exact occurrence covector and its positive EOCC evaluation are class-relative source results; finite acyclic interventions still admit a simultaneous atemporal factorization and hence do not prove ontic parent evolution.

The same downstream audit sharpens the rank-four handoff without promoting it into a measure identity. The retained CCOC connector is rank three on the relative observer–source sector, while OGAS places its contact row and the occurrence row on one signed local rank-four section. These data do not entail a full connector from $T(\mathbb{CP}^1_s \times \mathbb{CP}^1_b)$ to the body split $E_T \oplus E_S$, nor equality of their projective and body volume measures. If such a source-owned connector is additionally required to preserve the declared split and product measures, its common-line Jacobian is uniquely fixed to g_{FFL}^{-3}; there is then no extra temporal gain. Existence and physical occupation of that connector remain open. The common M4 SI scale is instead the metrological quotient of Eq. (32): one measured anchor selects a representative, but normalized morphology does not predict its decimal value.

6 Primitive readout families

Equation (78) gives the common explanation of the working primitive atlas:

Time.
Metric readout of positively ordered, observer-resolvable changes. Body orientation and clock form supply a continuous precursor; observer access and stable temporal resolution determine which transitions are ticks. Position and extent may change elapsed-time readout without changing the shared causal orientation. On the neutral regular leaf this construction reduces exactly to local GR proper time; a nonzero Tau-over-GR remainder is an open empirical claim, not a consequence of observer dependence alone.

Quantum.
Distinct accessible configurations can lie in one ordinary resolved class while a source-owned complex phase terminal retains relative holonomy and coherent composition. Loss alone is only coarse graining; Hilbert/Born, tensor and no-signalling structure require the additional positive state/effect, trace, composition and locality packet.

Gravity and geometry.
Effective geometric response of the accessible body stiffness, including Lorentz metric, causal cone, tidal response, stress coupling and null/Jacobi optics. A metric readout alone is not Einstein dynamics.

Matter.
A stable occupied localized body mode whose identity survives the observer quotient and admits compatible records and interactions. This role does not by itself derive mass, spin, charge or the Standard Model spectrum.

Radiation and photon.
An open/null body relation whose accessible ordered 4D image carries phase, polarization and interaction data. Parent "propagation" is not a stored trajectory; the atemporal body contains the compatible source–observer null relation.

Gauge and electromagnetic structure.
Compatible comparison of locally different presentations, with connection and holonomy retaining relational phase. Charged matter, Maxwell dynamics and interaction limits remain additional recovery obligations.

Energy and stress.
The observer-time generator descended from the common action unit, and the response of that action to the descended geometry. This row links time, matter, quantum and gravity and must be counted once.

The weak quantum notation $q = U_q^{M_\tau}(\rho_\tau(s))$ does not imply Hilbert space, interference or the Born rule. The MVP obtains trace form only after the positive state/effect assumptions A6; information loss or finite capacity alone gives coarse graining, not complex quantum mechanics [12, 6, 17, 3]. Quantum stress and semiclassical gravity share the common functional, but metric quantization, interacting UV QFT and a complete measurement ontology remain open.

For gravity, an obstruction is meaningful only as $\omega_g(r) = 0 \Leftrightarrow J_g U_g(r) N_g r$. A chart difference $J_g U_g(r) - r$ is secondary, and an empirical residual may not select the body. Dark-sector inference can mix gravitating content, baryonic remnants, modified response, morphology, path, observer and calibration effects. The conditional Einstein leaf does not choose among them or explain dark matter or dark energy.

A cosmological readout may be typed as

$$O_{\text{cosmo}} = U_{\text{cosmo}}(\rho_\tau(s), M_\tau^U, \gamma_{\text{path}}, \mathcal{O}_{\text{obs}}).$$

This warns against compressing expansion, distance, growth, lensing and CMB tensions into one unconstrained correction. A physical branch requires multi-probe covariance, source-native descriptors and controlled ΛCDM limits. Neither Λ_{ren} nor its numerical value is derived here.

Redshift, lensing, galaxy rotation and the CMB are composite readouts. They combine several primitive rows after one common observer-access stage. A dark-sector residual is a downstream mismatch against a complete frozen standard comparator, not an additional primitive readout and not a license to choose M_τ, A_O, terminal weights or quantizers from endpoint data.

7 Physical positioning and adjacent programs

Tau Core is neither anti-MOND nor anti-dark-matter. MOND-like laws may be effective limits, while lensing, CMB, BAO, clusters and growth still require joint source-frozen tests; this paper does not claim that collisionless matter is illusory. Relational clocks [18, 7], shape dynamics [11], semiclassical gravity [15], locally covariant QFT [4, 13, 14] and holographic error correction [1] are adjacent programs, not evidence for Tau ontology. The inverse-energy QOR form is a standard speed-limit structure [16, 2, 19]; only the independently reconstructed same-source body–Uhlmann identity is proposed as Tau-specific.

8 Falsifiability, objections, and next work

Tau Core becomes physics only through bounded source-first tasks. A branch must declare a noncircular response law and source-realizable body, descend through a quotient-safe readout, recover the appropriate known limit, and freeze at least one consequence that could fail before inspecting the endpoint. Negative results must demote a branch rather than authorize a new terminal or retuned kernel.

The principal failure modes are: no source-realizable parent object; a readout that is not well defined on its declared quotient; failure of the relevant GR, Newtonian, Born or cosmological limit; endpoint leakage; and rescue by an unconstrained post-hoc terminal. The corresponding status labels are *toy-only*, *phenomenological-only*, *blocked* and *rejected*. Toy-only means that a construction is internally consistent but not physically selected; phenomenological-only means that an effective fit exists without a parent derivation; blocked means that a named missing theorem or datum prevents promotion; rejected means that a frozen prediction, quotient condition, or known limit has failed. In particular, a coordinate declared null cannot later be used as an endpoint-selected correction without failing quotient safety.

The strongest reviewer objections are therefore legitimate tests, not matters of terminology. Without source-frozen constraints the parent is only a relabeling; without a physical realization law it is unobservable scaffolding; without preserved negative results the framework could explain anything. The displayed Einstein and Born leaves are conditional standard recovery, not first-principles derivations from the weak Tau spine. The QOR inverse-energy dependence is standard; only the independently reconstructed same-source identity $x_A = x_Q$ is Tau-specific. The once-counted ledger removes formal double counting only after microscopic ownership is supplied. Quantized or strongly fluctuating geometry and extreme semiclassical regimes remain outside the current theorem.

Structural novelty must take the form

$$\mathrm{Struct}_\tau(s, M_\tau) = 0 \Longrightarrow \mathrm{Constraint}(U_i^{M_\tau}, U_j^{M_\tau}, N_i, N_j),$$

or an upstream exclusion of a readout pattern. Merely adding a residual term to MOND, ΛCDM, GR or quantum theory does not qualify. At minimum a promoted branch must provide source declaration, body-first order, common ownership, readout codomain, null and relift policies, quotient safety, known limit, dimensional audit, a new discriminator and an explicit failure rule.

The next decisive task is the first open arrow

$$B_\tau^{\mathrm{phys}} + s_U \longrightarrow \Gamma_{\mathrm{MVP}} + \text{state/effect completion} + \text{observer carrier}.$$

It requires physical realization and occupation of the coherent Π_{phys} packet, occupied record support, the same-action A8b connector, universal-seed order compatibility, microscopic once-counting and record hold. A laboratory route must independently reconstruct body action, state distance, coherent/recovery evidence, energy spread and timing on one frozen carrier. No

platform is claimed to have completed that certificate. Superconducting qubits, trapped-ion or spin systems, metastable atomic levels and photon polarization are candidate carriers only, not asserted Tau realizations. Likewise, cross-readout residuals are diagnostics only after matched null, covariance and source-freeze controls; they are not by themselves validation statistics.

One conventional elastic-sheet control now makes a narrower architectural point concrete. A two-parameter gapped stiffness family can select a rank-two modal projector before any endpoint is read. Its Kato transport, reconstructed only from the frozen spectral projectors, agrees in the adiabatic limit with a second reconstruction obtained by propagating the complete elastic wave operator and removing only the source-known common dynamical phase. Fast driving and closure of the spectral gap destroy this agreement. Thus the body–occurrence–readout separation and its double-reconstruction test are realizable in standard mechanics; they are not peculiar to Tau Core. The sheet is an ordinary subsystem of the one physical body, its microphone and sensor records are terminal projections, and the laboratory time coordinate is a calibrated terminal clock. This control validates a method and supplies failure controls, but neither identifies the universe-level morphological body nor establishes physical occupation of a Tau parent law.

9 Global body selection and the remaining evidence obligation

The source-fidelity completion sharpens the global selection boundary. Split an admitted richer completion into the seed-generated coordinate x and a genuinely additional noncontractible coordinate (e). For the physically owned once-counted law

$$\mathcal{A}_{\mathrm{SF}}(x,e) = \mathcal{A}_{\mathrm{HC}}(x,e) + \mathcal{A}_{*} D(\widehat{\rho}_s \| \rho_{x,e}), \qquad \mathcal{A}_{*} > 0,$$

both terms are nonnegative and have a common zero at the faithful seed target. The reduced first variation in the extra sector therefore obeys

$$k_{\mathrm{eff}} = k - B^{\dagger} A^{-1} j = 0.$$

With positive Schur stiffness $S_E = C - B^{\dagger} A^{-1} B > 0$, the exact stationary solution is $e_{*} = S_E^{-1} k_{\mathrm{eff}} = 0$. A complete faithful characteristic evaluator makes the seed completion the unique minimum modulo the declared gauge relation. Thus this parent law does not occupy additional topology even when richer candidates are included before minimization.

This is an internal physical-action theorem, not evidence that Nature owns that global law. An independently sourced component zero-jet can preserve the local seed relations, Hessian and Hodge packet while reversing the global minimum. Consequently physical promotion requires either a deeper source derivation excluding such component data or a source-frozen empirical discriminator sensitive to them. Terminal agreement alone cannot discharge this obligation.

9.1 A concrete pointed-exterior realization of the finite closure

The abstract occupation boundary above now has one explicit finite-local realization. A full-rank incidence between oriented three-dimensional seed and axial carriers induces

$$\Lambda^2 J_s \oplus \Lambda^3 J_s : \Lambda^2 E_{\mathrm{src}} \oplus \Lambda^3 E_{\mathrm{src}} \longrightarrow \Lambda^2 E_{\mathrm{ax}} \oplus \Lambda^3 E_{\mathrm{ax}},$$

with ranks $3 + 1$. The two degrees inherit one Hodge metric and one action unit. A pointed exterior seed together with a faithful Hodge base and the source-fidelity/no-bypass law then occupies both degrees. Deleting the degree-two image costs three missing unit directions in the fidelity defect; deleting degree three costs one.

This result is stronger than merely postulating two independent closure edges: both arise from one seed incidence. It is nevertheless conditional on the enriched pointed-exterior parent

class. The narrow relation/order seed reduct does not entail that state-and-possibility grammar, and the existing top-form closure cannot linearly generate the axial pair closure because $\mathrm{Hom}_{S_4}(V_{\mathrm{ax}}, \Lambda^3 V_{\mathrm{ax}}) = 0$. Technical Paper VII-B owns the detailed theorem and controls. Paper I imports only the architectural consequence and retains unrestricted physical base–seed ownership as open.

9.2 Hodge–Clifford root freedom inside the occupied class

The same global completion also removes one apparently independent terminal choice. Let the charged carrier factor as $\mathcal{A}_{\mathrm{ch}} = M_4(\mathbb{C}) \otimes \mathcal{A}_{\mathcal{M}}$, and let the charged restriction $K_s \geq 0$ of the complete source Hodge operator commute with the full geometric Clifford factor:

$$[K_s, X \otimes I] = 0 \qquad \forall X \in M_4(\mathbb{C}).$$

The matrix-factor commutant then gives $K_s = I_4 \otimes K_{\mathcal{M}}$. For the standard full-support spectral occupation,

$$\rho_s = \frac{e^{-K_s/\mathcal{A}_*}}{\mathrm{Tr}\, e^{-K_s/\mathcal{A}_*}} = \frac{I_4}{4} \otimes \rho_{\mathcal{M}}.$$

Hence the geometric functional is tracial, no marked spin root survives, and the photon and observer sectors inherit one cone without an added cone gain. This is an exact consequence inside the seed-fixed Hodge-generated completion class, not a proof that the unrestricted physical base–seed law occupies that class. A faithful exponential state alone is insufficient: a rooted faithful state is also exponential, but its logarithmic generator fails the stated Clifford naturality. Likewise, componentwise source zero-jets can preserve the entire local Hodge packet while changing global occupation. The remaining obligation is therefore class membership or a source-frozen cone-split test, not another root-deletion postulate.

9.3 Inherited full-4D standard-excess compiler

The observer-specific full-coframe branch now supplies a finite downstream compiler without changing the parent ontology of this paper. On one frozen visible–unresolved split, the same positive parent two-jet has blocks K_{HH}, C, K_{VV}, and minimum-lift descent gives

$$K_O^{\mathrm{quot}} = K_{HH} - C K_{VV}^{-1} C^\dagger.$$

The object that can be compared with a standard physical leaf is not the positive Schur term alone but

$$E_K = (K_{HH} - K_{\mathrm{std}}) - C K_{VV}^{-1} C^\dagger. \tag{81}$$

A graph-normal auxiliary completion can have $C \neq 0$ while $E_K = 0$. Hence a shared parent mixed Hessian is an internal coupling datum, not by itself a Tau-over-standard anomaly. After one regular coframe pushes E_K to the 4D metric excess, frozen terminal maps may be stacked only with their full cross-terminal covariance and after exact duplicate directions are quotiented. This is an inherited scoring-compiler theorem. Paper I does not claim that the physical parent occupies a nonzero packet or that any existing endpoint has produced such an excess.

9.4 Common quantum–gravity transport and action-unit boundary

Let one faithful occupied source quotient carry a positive generator K_*, and let W_Q, W_G be faithful source-owned realizations of its quantum and gravity terminal images. On the occupied ranges the only source-compatible transfer is

$$T_{G \leftarrow Q} = W_G W_Q^{-1}, \qquad T_{G \leftarrow Q} K_Q = K_G T_{G \leftarrow Q}, \tag{82}$$

where $K_e = W_e K_* W_e^{-1}$. If both terminal metrics pull back to the same source Hessian,

$$W_Q^\dagger h_Q W_Q = W_G^\dagger h_G W_G = K_*,$$

then $T_{G\leftarrow Q}^\dagger h_G T_{G\leftarrow Q} = h_Q$. Thus a common parent does not permit an arbitrary fitted rescaling after the occupied source maps have been fixed.

Inside the declared faithful SHR–PSAL–QPHASE completion, standard quantum phase calibration fixes the common action unit to

$$\mathcal{A}_* = \frac{S_0}{2\pi} = \hbar, \qquad \mathcal{A}_* = C_G \ell_H^4 = \frac{\ell_H^2}{6\kappa}. \tag{83}$$

Consequently $\ell_H^2 = 48\pi\hbar G/c^3$ in the stated action convention. This is a conditional cross-terminal normalization lock, not a derivation of the SI value of $\hbar$ or proof that Nature occupies the completion. Nor do separate published quantum and gravity marginals reconstruct W_Q, W_G or their joint source covariance; those require a common latent source model or common-event data.

The residual fanout freedom also closes inside a sharper enriched packet. If the occupied common source sector is absolutely irreducible, its quantum and gravity copies occur with multiplicity one, and the terminal maps are source equivariant, Schur rigidity makes every nonzero map unique up to a scalar. Cyclic no-bypass occupation excludes the zero map, the common pullback metric fixes the scalar magnitude, and a source-owned oriented anchor fixes its sign. Hence (82) is then the unique source-natural isometry. The zero-map, repeated-copy and unfixed-anchor controls show why each premise is necessary. This is an internal conditional closure; it does not prove that the unrestricted physical base–seed law occupies this sector.

One endpoint-blind Schur-neutral relation-defect square can generate any supplied positive response Hessian and mixed source jet while reducing exactly to the original seed–body action after the post-body variable is minimized. It therefore closes the scalar action-form search without back-selection. However, orthogonally rotated terminal projectors leave that scalar action unchanged while altering the named terminal responses. The minimal sufficient object in this declared class is consequently a complete graded source characteristic containing both the joint jet and its source-owned role representation. Its physical ownership remains open.

The pointed exterior result reduces this remaining fanout packet further. On an onto, no-bypass CAR orbit the empty root is cyclic, the represented source algebra is a full matrix algebra, and its physical module is absolutely irreducible with no spectator copy. Tensoring this module with one source-owned oriented two-role plane gives exactly one quantum copy and one gravity copy; a common product metric then supplies the remaining hypotheses of the Schur closure above. Conversely, without that typed role plane a $U(2)$ rotation preserves the untyped scalar action and metric while changing the named projectors; zero-leg and duplicate-copy controls also survive. Thus the unresolved physical input is one nonzero oriented Q/G incidence in the complete graded source representation, not six independent fanout coefficients. Inside the declared enriched P3 plus Tau-local composition class, that incidence is not an additional connector: the two source-owned nonzero ordered cotangent ports supply its metric-dual Q/G lines, and local source-copy factorization supplies one copy per role. A hidden-diagonal control retains full Q and G marginals while halving the joint rank, proving that the factorization premise is essential. The paper does not derive that the physical base–seed law jointly realizes the enriched port packet and this Tau-local factorization.

The remaining two premises are jointly realized by the previously constructed component-local enriched action $\widehat{\Gamma}_{\mathrm{CLP3}} = \sum_C \widehat{\Gamma}_C$. Its nonnegative, once-counted component laws have attainable zeros; every global minimizer therefore occupies both the enriched P3 port packet and LC1–LC4. Composing this result with the pointed-CAR and Schur theorems gives the unique normalized isometric Q/G transport, and the faithful phase specialization gives $\mathcal{A}_* = \hbar$, without a further connector or gain. This closes the conditional completion-to-fanout chain. It

does not show that Nature selects Γ_{CLP3}: hidden-global-relation and missing-port completions have the same narrow upstream reduct and fail the conclusion.

More precisely, the physical target is not one preferred off-shell scalar formula. The fanout is invariant over the class of nonnegative completions with the same occupied minimum set, true-normal physical Hessian in the common unit, and typed port/root/phase packet. Indeed, $\Gamma_\lambda = \Gamma_{\mathrm{CLP3}} + \lambda\Gamma_{\mathrm{CLP3}}^2$ has the same minimum, first variation and physical Hessian for every $\lambda > 0$ while differing off shell. Nature selection therefore means membership in this completion class; the hidden-global and missing-port controls lie outside it.

10 What is not proved

For clarity, we list the main non-results.

- The paper does not prove that the Tau substrate, universal seed, or complete universe-scale morphological body exists physically.
- It does not derive the adopted enriched body action and completion uniquely from the physical base–seed source. The current narrow source reduct provably admits both the primary source-irredundant completion and an inequivalent energy-sharing ownership completion.
- The Hodge–Clifford argument proves root freedom and a common cone only inside the seed-fixed Hodge-generated completion class. It does not prove that the actual physical base–seed system belongs to that class.
- It does not derive the microscopic ownership partition or full nonlinear action whose integration yields the once-counted effective functional in Eq. (21).
- It does not predict G, Λ, $\hbar$, gauge couplings, particle masses, or other constants numerically from parent data. A downstream scalar-15 specialization fixes the class-relative ratio $M_*/E_P = 1/\sqrt{24\pi}$ and obtains its SI value from one CODATA anchor; this is neither a Paper-I result nor a Tau prediction of G. The equality $\mathcal{A}_* = \hbar$ above is a standard-limit calibration inside one declared common-action completion, not a parent-level numerical prediction.
- Conditional Einstein and Born recovery is not an unconditional derivation of GR or quantum mechanics from the minimal Tau base.
- It does not provide ultraviolet quantum gravity, nonperturbative interacting QFT, a complete measurement ontology, or a metric-quantization theorem.
- It does not derive the full Standard Model, its spectrum, generations, mixing, or neutrino sector.
- The QOR inequality is not a universal time atom or an extra delay. The support theorem and mixed-state realization prove that the common-Gram completion is mathematically nonempty, but the physical parent has not been shown to occupy the record instrument or select $V_{\mathrm{phys}} = V_{r(x_O)}$. No laboratory carrier has validated the same-source identity $x_A = x_Q$.
- The dagger-Frobenius unfolding theorem and its once-counted traced source realization supply a normalized record-copy mechanism inside the enriched two-leg completion. The source-irredundant branch is selected internally, but the present narrow parent action does not select that enriched completion from the unrestricted physical base–seed parent law. Once an occupied complete record edge of that enriched packet is supplied, finite A8b is derived on its complete path orbit; the paper does not prove physical-parent or laboratory occupation of the orbit, nor that it covers arbitrary quantum states.
- The complete-edge theorem does not cover generic irreversible CP, decohering or rank-reducing record processes. Exact A8b for such processes would require a separately proved recovery-correctable or equivalent nonisomorphic extension.

- The finite joint-state and colimit reductions do not prove physical occupation of the coherent source homomorphism Π_{phys}; all proper marginals remain insufficient.
- It does not eliminate or explain dark matter or dark energy.
- No empirical result presented in this manuscript validates the Tau ontology.

These limitations are not footnotes. They are part of the claim. A foundation paper that hides its blockers is not useful. The goal is to make the blockers visible enough that later work can attack them one at a time.

11 Finite-local occupied-body support

For a finite smooth local extension, split the retained body coordinate x from a genuinely additional coordinate e, with once-counted Hessian and load

$$H = \begin{pmatrix} A & B \\ B^\dagger & C \end{pmatrix}, \qquad \ell = \begin{pmatrix} j \\ k \end{pmatrix}.$$

If $A > 0$ and $H > 0$, exact Schur elimination gives

$$e_* = (C - B^\dagger A^{-1} B)^{-1}(k - B^\dagger A^{-1} j). \tag{84}$$

Thus positive extra stiffness alone does not remove a sector. A positive, seed-unloaded and decoupled sector has $B = k = 0$ and $e_* = 0$; a stable loaded or cross-coupled sector is part of generated body support. This closes finite C^2 local occupied-support completeness in the declared working class. It does not prove ambient global, topological or infinite-dimensional exhaustivity, or unrestricted realization of the MVP packet.

The upstream parent-realization result supplies one precise operational convention. After quotienting directions annihilated by every physically owned action, coupling and terminal, the enriched one-source premise identifies the occupied physical parent with the universal-seed-generated closure. An inert ambient direct summand is then only presentation surplus; a physically non-null direction at fixed complete base–seed key is a second source and falsifies the one-seed architecture. Paper IV owns the proof. This paper adopts the result only conditionally and does not claim that old-bare Tau, 4D recovery or observation establishes the one-source premise.

Unified source-selection and Nature-test handoff

Foundation Paper IV now owns a sharper conditional result, denoted PD-USS1. If one source-irredundant staged functional has a unique body minimum, an isolated source-owned spectral carrier, a grading intertwiner, a unique positive-defect reversible transport law, associated terminal representations, no bypass and global-minimum actualization, then it selects one state family, one transport structure and one common terminal carrier modulo gauge. The grading, smooth transport and actualization premises are not jointly derived or physically occupied by the present Tau source. Moreover PD-USS-N1 proves that even this complete selection does not force a standard-leaf-orthogonal signal. Technical Paper VII-B owns the first terminal falsifier and the updated public-data census: no currently audited packet among 31 candidates supplies the required independently typed rank-two pair. Thus the architecture has a conditional selector and an executable falsification target, but no Nature-level occupation claim.

Inherited actional closure of the local physical quotient

Foundation Paper IV now supplies ACT-PCLOSE1, the precise conditional bridge from one ordered Parent-action packet to all declared readouts. After the seed–body solution is selected

and frozen, regular post-body stationarity gives

$$D_S Z_* = -H_Z^{-1} B_{ZS}, \qquad O_i = Q_{OS,i} \circ T_i \circ P_i \circ \Xi(Z_*).$$

The stacked terminals reconstruct the declared local physical quotient iff $\ker J_{\mathrm{term}} = \mathcal{N}_{\mathrm{phys}}$. This sharpens the architecture: the named time, quantum, gravity and other outputs are typed descendants of one response rather than independent fits. It does not prove global source exhaustivity, Nature occupation, Standard-Model content, absolute units, or a nonzero Tau-specific signal. A stable terminal-blind source sector is the exact countermodel to any stronger inference.

The post-body active solder needs no additional fitted matrix when it is read from the same frozen action. With response Hessian H and mixed Hessian B, implicit stationarity gives

$$\Gamma = D_Y Z_* = -H^{-1} B. \tag{85}$$

For the inherited SBTCL106 Bregman action one has $H = G_{OS}$, $B = -G_{OS} J_{OS}$, hence $\Gamma = J_{OS}$ and $C - B^* H^{-1} B = 0$. This is an internal conditional reduction, not a physical source-selection result: descent through a proposed active source variable still requires $p_s J_{OS}|_{\ker DF} = 0$, and neither its source ownership nor nonzero physical occupation follows from positivity or Schur neutrality.

12 Conditional Graph/Hilbert representation and forward-only placement

A graph of complex compatibilities is admissible in this paper only after the source, body and observer–source descent have supplied an algebra $\mathcal{A}_{OS}$, a positive state ω_{OS} and non-null probes x_a. The normalized GNS overlaps

$$G^{OS}_{ab} = \frac{\omega_{OS}(x_a^* x_b)}{\sqrt{\omega_{OS}(x_a^* x_a)\omega_{OS}(x_b^* x_b)}} \tag{86}$$

then define a Hermitian positive-semidefinite matrix with $G^{OS}_{aa} = 1$. Conversely, every finite Hermitian positive-semidefinite unit-diagonal matrix is a Gram matrix of unit rays, with minimal Hilbert dimension $\operatorname{rank} G^{OS}$. This is standard finite-dimensional mathematics, not a new Parent-selection theorem.

The exact forward placement is

$$B_\tau^{\mathrm{base}} + \sigma_U \longrightarrow M_\tau \longrightarrow (\mathcal{A}_{OS}, \omega_{OS}, \{x_a\}) \longrightarrow G^{OS} \longrightarrow \text{typed terminal readouts.} \tag{87}$$

The reverse arrow is invalid. Indeed, both I_n and $\mathbf{1}\mathbf{1}^*$ are unit-diagonal positive Gram matrices, but their ranks, correlations and graph structures are maximally different. Gram positivity therefore does not select the universal seed, stabilize the body, identify an observer, or prove Nature occupation. Low modes, projective phases and graph clusters remain conditional diagnostics until an independently source-owned action and terminal map give them physical roles. This correction preserves the paper's architecture: the Hilbert graph is a possible representation of an already derived shared readout, not a second Parent ontology and not a route for inferring the body from its terminals.

13 Expansion-conditioned shared-spectrum specialization

The ECTR packet provides a conditional specialization of the architecture, not a new upstream ontology. Let the frozen seed–body stage supply a positive spatial carrier (E_3, h_S) and body symbol $J_{M,S}$, so that

$$A_{0,M} = J_{M,S}^* h_S J_{M,S} > 0. \tag{88}$$

If a separately supplied, source-normalized Legendre–Riesz coevaluation is occupied after body stabilization, its dual source Hessian can satisfy

$$\boxed{G = \frac{K_{\mathrm{src}}}{2A_*} = k_{c,O}^2 A_{0,M}^{-1}.} \tag{89}$$

The descended coframe and Q leg are then two different maps of this same operator packet,

$$q_{a,M} = a^2 A_{0,M}^{-1}, \qquad F_{Q,O}(a) = (I + a^2 G)^{-1/2}. \tag{90}$$

The causal reading is a common-source fork. The frozen source packet supplies both the metric-active scale a_O and the expansion-conditioned access law; a_O may therefore modulate terminal dominance after that law and the typed exits are fixed. Observed 4D expansion does not select G, $k_{c,O}$, the stable observer–source quantizer or quantum/gravity terminal semantics. On a positively calibrated occupied access ray the determinant loss $\Gamma_{U,O} = -d_{\tau_O} \log\det F_{Q,O}$ and the scale elasticity $\mathcal{E}_Q = -\partial_{\log a} \log\det F_{Q,O}$ obey

$$\boxed{H_O = \frac{\dot a}{a} = \frac{\Gamma_{U,O}}{\mathcal{E}_Q(a)}}. \tag{91}$$

For the Q–metric lock both factors are positive when the R1 access rate is positive; zero access gives the exact static control. The sign glues only over compatible comoving observers on one oriented, time-oriented globally hyperbolic leaf. It is not a statement about arbitrary observers.

This result sharpens rather than closes the parent-realization gate. The same seed, body and coframe admit different source coevaluations, and the same spectral packet admits zero access. Hence the universal implication is conditional: a Q–metric lock with positive occupied loss implies expansion. It does not say that every Tau-admissible universe expands, that our universe occupies this class, or that acceleration, dark energy or a particular $H(z)$ has been derived. Inferring the upstream lock from observed expansion would reverse the generative arrows of this paper.

14 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS}.

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. The occupied observer–source context is relational instance data, not a uniquely selected observer identity inferred from the universal seed alone. Physical source realization and record preparation remain separate from conditional closure.

15 Finite source-to-readout implementation and its scope

The current finite laboratory is a reproducible implementation, not a new Nature-selection theorem. The executable lab now instantiates the forward order: supplied finite source genes, stabilized body, frozen-body observer response, stable acceptance cells, then terminal diagnostics. It searches finite components, not independently seeded galaxies or complete universes. The supplied genes and downstream laws retain their separate ownership obligations.

Ownership and reproducibility. The seed–body companion owns the finite minimum/Hessian example, the body–readout companion its ordered access and contact reduction, and the terminal companion its conditional propagation boundary. The physical-source papers retain the source-selection obligation. Code, declared priors, adverse controls and provenance are available at https://github.com/tau-core-research/tau-core-theory/tree/main/experiments/seed_evolution_lab. Numerical tests are internal evidence only; no published empirical score is changed.

16 Source identification and recovery

The current observer audit separates a conditional forward construction from physical source selection. A frozen body, occupied observer incidence, preparation, complete instrument and stable resolution can determine records without selecting those inputs from Nature. Equal effects do not fix postmeasurement states, and symmetry within a chosen interaction family does not exclude other symmetric couplings. The occupied observer–source context is relational instance data, not a uniquely selected observer identity inferred from the universal seed alone. Physical source realization and record preparation remain separate from conditional closure.

What remains of selection. The positive relation-energy grammar admits different observer interactions with identical reduced symmetries or spectra but different instruments. The complete interaction may fix its forward output uniquely; a reduced packet does not. This is a completion-selection boundary, not a requirement that the old minimal base alone determine every coefficient. The universe-specific seed hypothesis must retain independently justified base laws and irreducible local observer incidence. The original EOCC gradient prescription and an inertial reversible-plus-loss extension likewise require distinct physical ownership; shared equilibrium does not identify their actual evolution.

Dependency and claim ownership. Foundation VI and the body–readout technical companion own the finite instrument/access proof; Foundation IV and TD-0 retain physical

source selection; Foundation V, the temporal synthesis and seed–body companion own the gradient/inertial distinction. This paper inherits only the stated specialization. Canonical assumptions, countermodels and calibration audits are in https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_rqm_instrument_to_record_bridge_001.md. The temporal source audit is https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_core_temporal_seed_base_morphogenesis_branch_001.md, sections 628–630. No physical seed, Nature occupation or Tau-exclusive observation is established by this update. Existing empirical endpoint definitions and scores are unchanged.

Imported result (CPR1). Paper 4, *Full Derivation Ledger for the Branch A Higgs Module*, section "Conditional source preparation and time-resolved access, Theorem CPR1(a–c), v1.0 (2026-09-28), is the sole proof owner. This is a conditional example of preparation and joint access after body stabilization, not a derivation of the occupied observer–source context. It neither adds a dynamical memory to the parent nor selects an operational observer from the universal seed. Physical preparation, detector/clock calibration and stable Q_{OS} remain assumptions; no Tau-exclusive observation follows.

Current source/readout status (9 October 2026). Foundation VI, OIR1(a–c), v1.0 ("Output statistics, surviving-source instruments and apparatus scope") and Paper 4 full derivation, MSI1, v1.0 ("Source work, bilinear mass pairing and independent selection"), both dated 9 October 2026, own the proofs. Body-resident ordering/compatibility has observer-indexed lossy realizations; the legacy word channel does not introduce a separate substance or a separately seeded body for each apparatus. OIR1 distinguishes records from the complete operation. It supplies no physical selection of the universal seed or occupied observer. These are conditional mathematics and standard-compatible controls, not Nature occupation or Tau-exclusive evidence. Stable Q_{OS} and physical calibration remain open; empirical scores are unchanged.

17 Conclusion

Tau Core is presented as an atemporal morphological readout framework with one frozen conditional completion. A stabilized body is solved before terminal variation, and one once-counted functional supplies compatible conditional semiclassical-Einstein/optics and Schrödinger–Born leaves. This is an auditable architecture, not a theorem that the physical base–seed source selects it.

The proposed finite discriminator is the independently reconstructed same-source identity

$$x_A = \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*} = x_Q = d^2_{U,\mathrm{chord}}(\rho_0, \rho_1),$$

which turns the standard quantum speed limit into the QOR bound. The novelty claim concerns this body–Uhlmann bridge, not inverse-energy timing and not a universal time atom.

Inside the declared enriched packet, a complete source-owned record edge and primitive extensivity are sufficient for exact finite A8b. Pathwise descent retains descriptor-visible holonomy, and commuting records do not identify reversibility. The strengthened FOC-7 certificate therefore requires a coherent witness or certified recovery in addition to independent action, tomography and timing measurements.

The decisive blocker remains unrestricted parent-law realization and occupation of the common source packet. A deeper source theorem or a successful source-frozen FOC-7 test could promote this completion; a reliable violation would reject it. Adding unconstrained terminals after such a failure would abandon, not repair, the minimum-viable theory tested here.

The later quantum-terminal analysis sharpens this blocker without changing the architecture. A normalized disjoint-source valuation is sufficient to make the once-counted parent action

additive on disconnected occupied source components and hence to remove their mixed Hessian block. Conversely, once-counting, positivity and symmetry alone permit a hidden positive cross-component term. Operational factorization lifts back to exact parent additivity only when the *joint* primitive-readout descent is faithful on such cross-component defects. Since the currently known physical terminal maps do not yet form one complete source-frozen stacked Jacobian, this faithfulness and therefore unrestricted physical parent additivity remain open.

The downstream scalar-15/type-II completion illustrates the same claim discipline at a more specialized level. Recovery-complete source generation can close ownership inside a declared branch, a common Planck terminal can calibrate its mass scale, and an exact functional flow can close its renormalization law. None of these statements promotes branch occupation to unrestricted parent-law realization, and none changes the foundation-level open arrow frozen here.

Within the series, the handoff is explicit: this paper freezes the architecture and conditional physical packet; Paper II asks which finite source data can identify one joint realization; Paper III asks whether that identified information survives reversible or recoverable record transport. Neither handoff closes the shared parent-realization arrow.

The atemporal–temporal relation is therefore no longer posed as two rival fundamental theories. The atemporal carrier architecture is the encompassing kinematical layer; a temporal completion adds an integrable order/actuality structure on a temporalizable carrier. EOCC conditionally actualizes that structure inside one enriched source class, while the finite interventional simulation theorem prevents its operational success from being promoted into a proof of literal Parent traversal. Nature-level temporal occupation, absolute proto-time calibration and observer-clock recovery remain distinct open claims.

A Mixed-state distance and QFI conventions

To avoid the common squared-versus-root fidelity ambiguity, this manuscript uses the *squared* Uhlmann fidelity

$$F(\rho,\sigma)=\left[\operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right]^2, \qquad 0\le F\le 1. \tag{92}$$

The root fidelity is $f(\rho,\sigma)=\sqrt{F(\rho,\sigma)}$. The Bures angle used in Eq. (42) is therefore

$$D_B(\rho,\sigma)=\arccos f(\rho,\sigma)=\arccos\sqrt{F(\rho,\sigma)}. \tag{93}$$

For pure states, $F=|\langle\psi|\varphi\rangle|^2$, so $D_B=\arccos|\langle\psi|\varphi\rangle|$, the Fubini–Study angle.

For a differentiable one-parameter mixed-state path, the infinitesimal Bures line element and quantum Fisher information $\mathcal{F}_Q$ satisfy

$$ds_B^2=\frac{1}{4}\mathcal{F}_Q(t)\,dt^2. \tag{94}$$

For unitary evolution generated by a self-adjoint H,

$$\mathcal{F}_Q(t)\le\frac{4\Delta H(t)^2}{\hbar^2}, \qquad \frac{ds_B}{dt}\le\frac{\Delta H(t)}{\hbar}, \tag{95}$$

which integrates to Eq. (45) [19]. A nonunitary channel must instead use its actual QFI path speed; Eq. (95) is not assumed for it.

The common-Gram completion fixes the notation

$$x_O=\frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*}=\epsilon_O^2=2(1-\sqrt{F}), \qquad D_Q=2\arcsin(\epsilon_O/2).$$

Hence the linear inequality $D_Q\ge\epsilon_O$ is a conservative corollary, with equality only in the zero-separation limit. The exact state-distance quantum speed limit is the first inequality in Eq. (46).

B Conditional finite source–state result chain

The 10 November 2026 ET/CAR record supplies a bounded downstream example, not a parent-source selection or empirical cosmological result. Foundation VII-A owns the detailed response presentation; its HSL section and compact receipt retain the numerical provenance. The inherited stages have different logical status:

1. HRZ/HSA derive the bosonic Hessian and the once-summed metric/scalar mean constraints from the same source functional. The original zero-derivative reference is not a saddle; positive volume alone does not repair it. This does not exclude the HORIZ preparation family.

2. HSD/HSE solve regular initial jets for that original preparation at $k = 16, 32$ and check all three mean Euler equations and Hamilton-constraint propagation. At 90-digit precision the checked residuals are below 4×10^{-85}. A frozen-state derivative control fails.

3. HSF evolves the background and occupied frame together on $0 \leq t \leq 0.02$, without instantaneous state refill or projection. RK4 refinement gives a ratio 15.979; the finest lapse residual is 4.29×10^{-10}. Freezing the state instead gives a residual about 4.58. HSG independently replays every fine-grid step, with discrepancy below 1.4×10^{-17}. The original numerical-backend warnings remain qualified, not explained away. HSH tests common-constraint regularity at the saved nodes, not a rigorous between-node or global stability bound.

4. HSI/HSJ retain nonzero common-source covariance and mandatory cross terms in the same transported Slater state. A vanishing mean constraint does not imply vanishing quantum noise. HSI's coarse-grid mean-binding check failed its unchanged tolerance; the separate finest-grid guard passes without erasing that failure.

5. HSL transports the full normal16 carrier on this same background and evaluates all nine retarded source blocks at three fixed times. Eight internal checks pass, with refinement ratio 16.4278 and state-transport discrepancy below 10^{-15}. This is not a continuum error certificate. The response formula retains the original preparation bracket, upper endpoint and local contacts; numerical contact/pulse assembly and the common metric Volterra solution remain open.

The assumptions are a finite homogeneous normal Hamiltonian, the original HORIZ state prescription, one source functional and original endpoint maps. An alternative admissible state can change covariance and retarded expectation. Neither normal finite evolution nor off-diagonal response establishes physical particle creation. Spatial and renormalized quantum–gravitational closure, physical state and Q_{OS} selection, general/all-state/UV stability, GR/UV completion and cosmological validation remain open.

The common-action recovery discussion therefore has a concrete finite-model example, not a new renormalized recovery theorem.

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